

Figure Used in this Manuscript

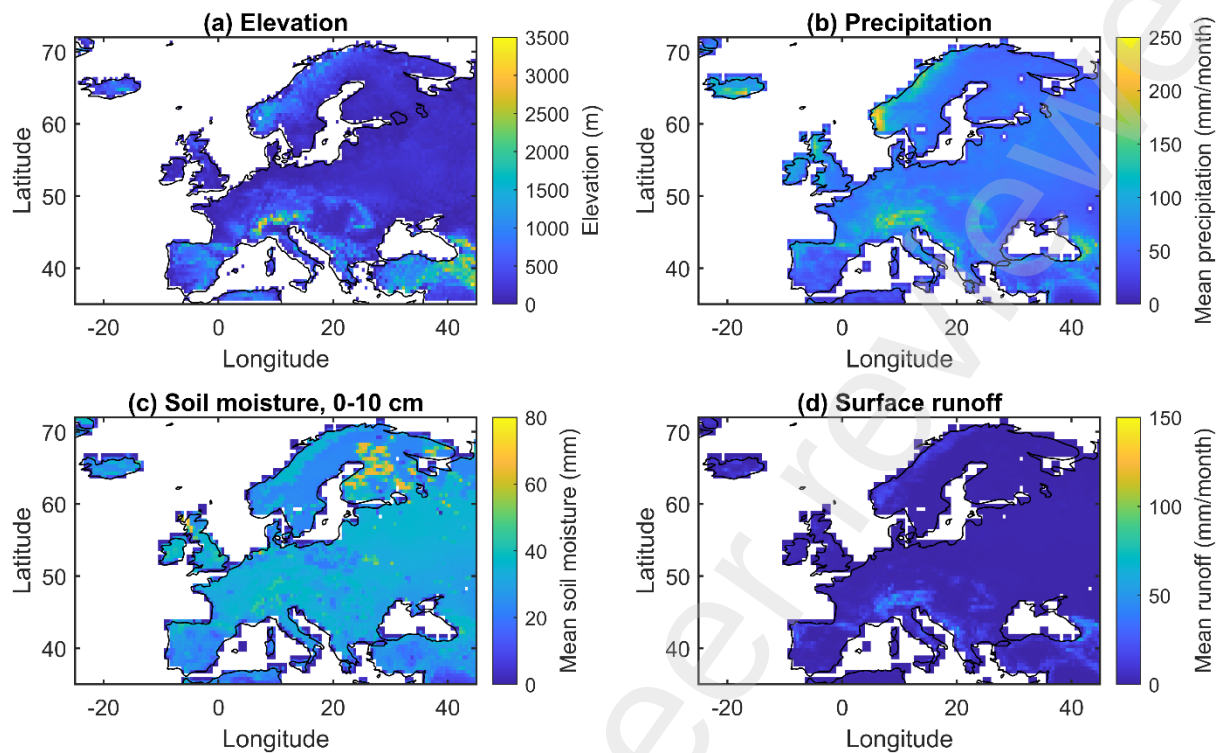


Figure 1. European study domain and spatial distribution of elevation and ERA5-Land reference precipitation, surface soil moisture and surface runoff during 1980–2014.

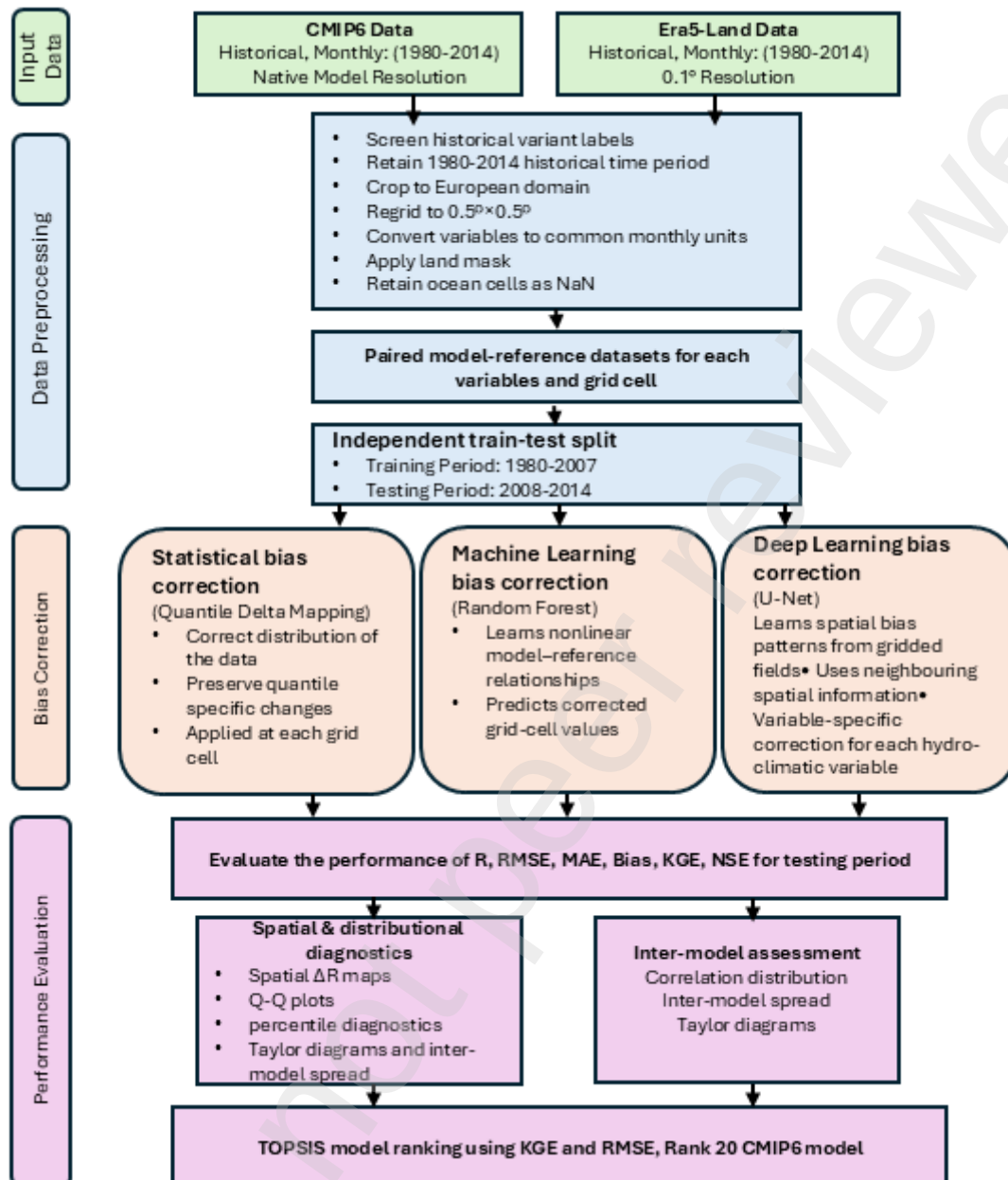


Figure 2. Overall workflow for preprocessing, bias correction and performance evaluation of the ERA5-Land and CMIP6 datasets.

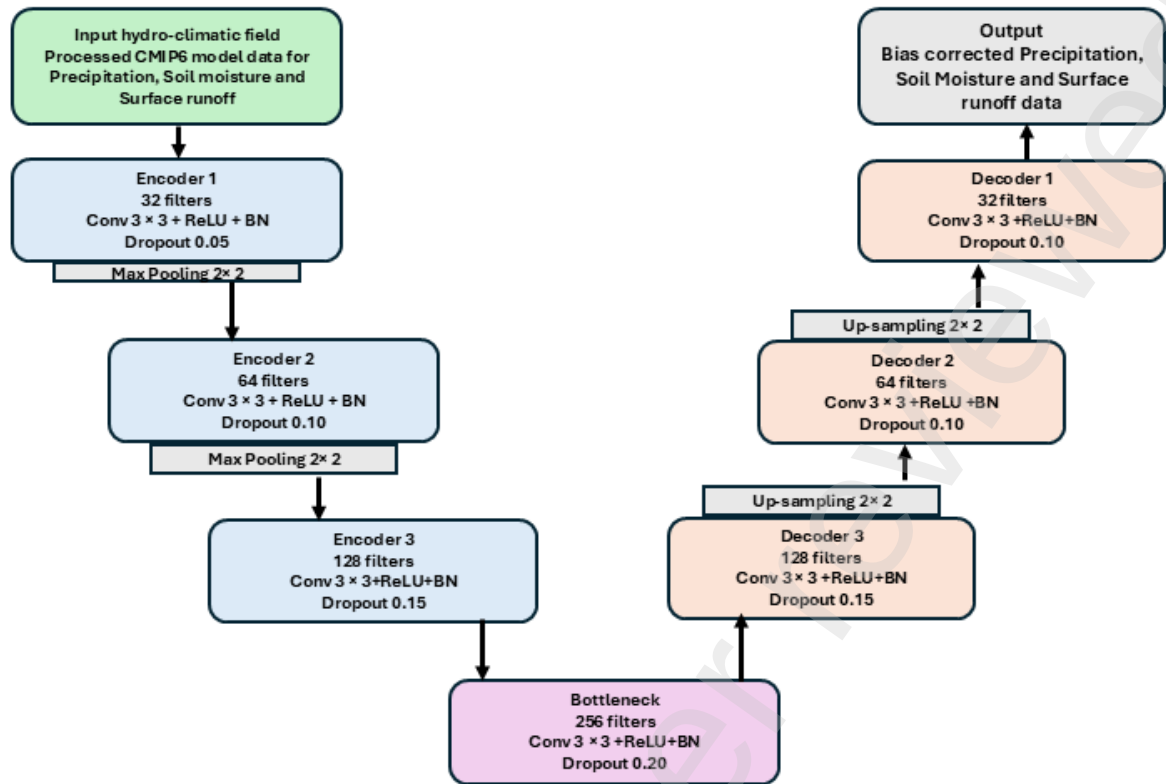


Figure 3. U-Net encoder-decoder architecture used for spatial bias correction of CMIP6 hydro-climatic fields.

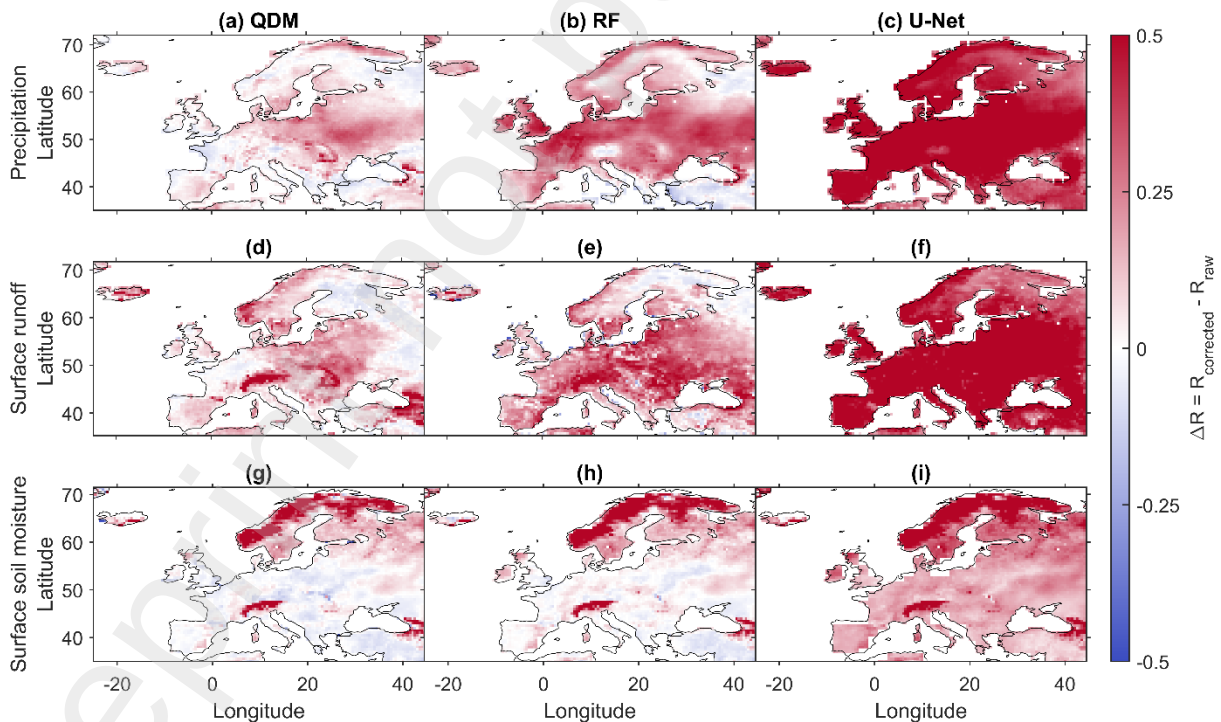


Figure 4. Spatial changes in temporal correlation following bias correction for precipitation, surface soil moisture and surface runoff during the independent testing period. At each valid European land grid cell, ΔR was calculated as $R_{\text{corrected}} - R_{\text{raw}}$ using the monthly time series. Positive values indicate increased correlation with ERA5-Land, whereas negative values indicate reduced correlation.

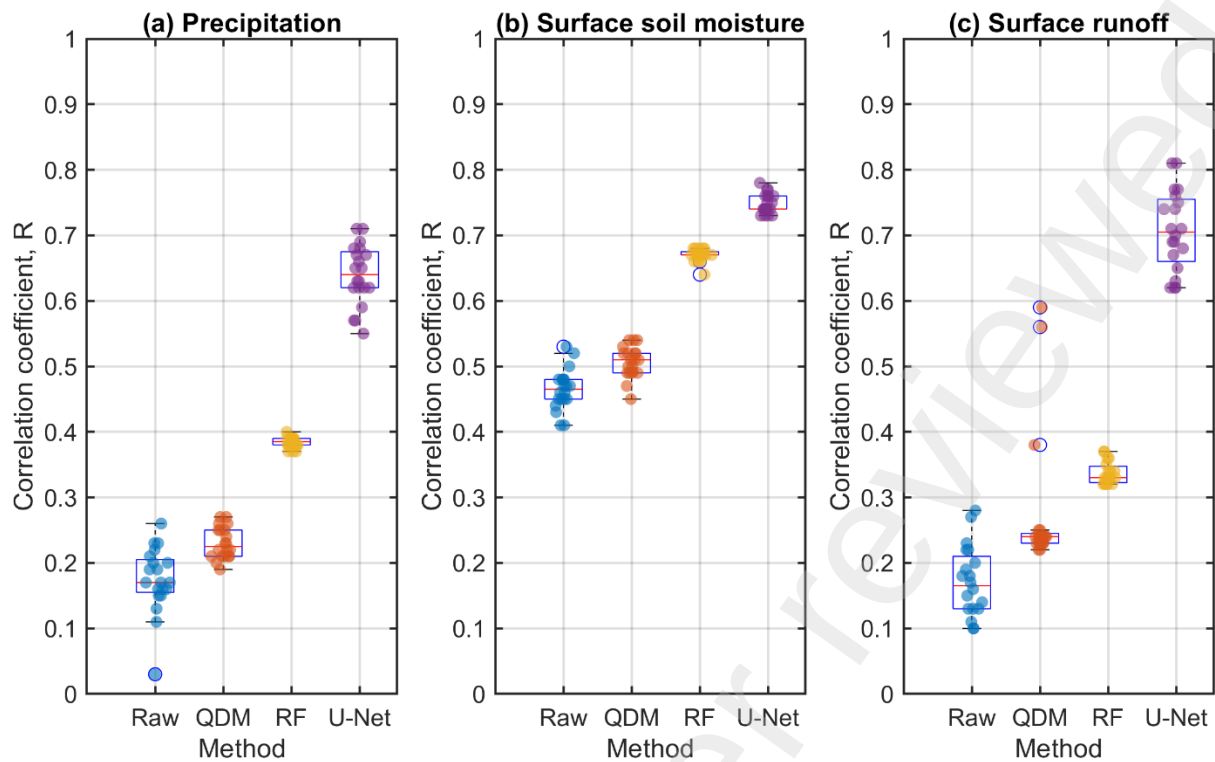


Figure 5. Distribution of temporal correlation coefficients across the 20 CMIP6 models for precipitation, surface soil moisture and surface runoff during the testing period. Each point represents an individual CMIP6 model.

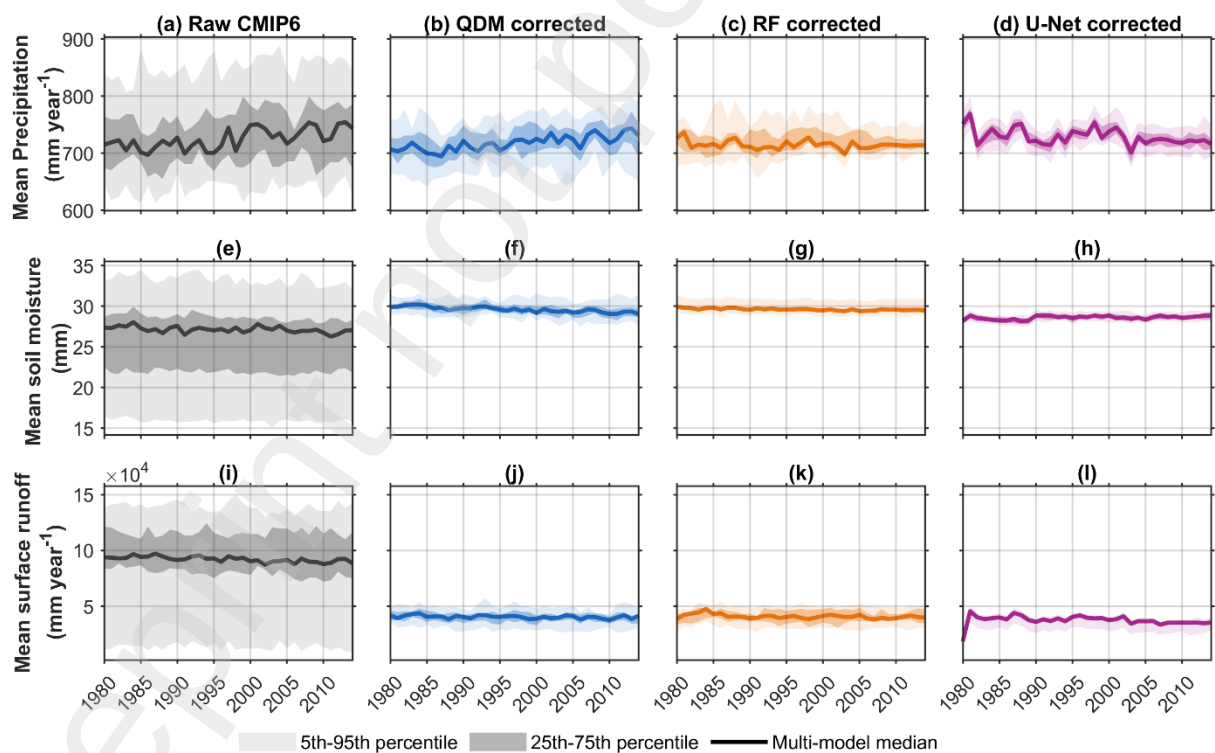


Figure 6. Temporal inter-model spread of precipitation, surface soil moisture and surface runoff during 1980–2014 for the raw, QDM-, RF- and U-Net-corrected simulations. The light and dark shaded areas represent the 5th–95th and 25th–75th percentile ranges, respectively, and the solid line represents the multi-model median.

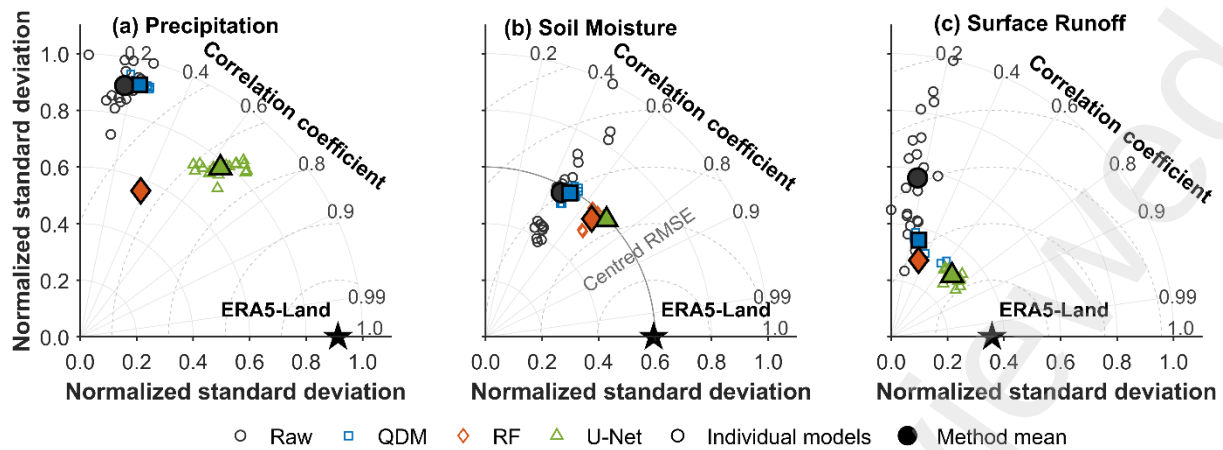


Figure 7. Taylor diagrams comparing the temporal performance of the raw, QDM-, RF- and U-Net-corrected CMIP6 simulations with ERA5-Land during the independent testing period. The diagrams summarize temporal correlation, normalized standard deviation and centred RMSE for precipitation, surface soil moisture and surface runoff.

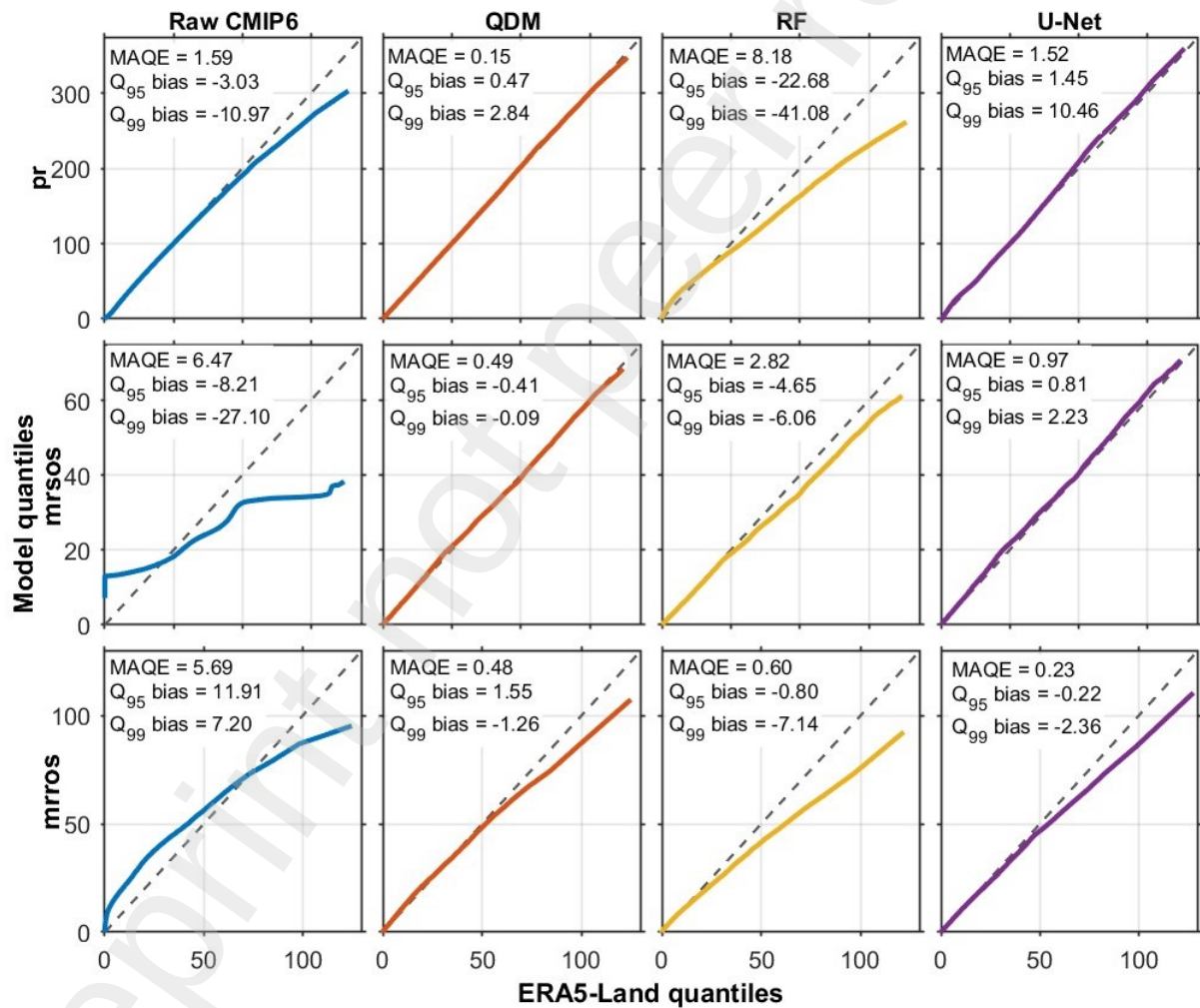


Figure 8. Q-Q comparison of raw, QDM-, RF- and U-Net-corrected precipitation, surface soil moisture and surface runoff against ERA5-Land during the independent testing period. The dashed 1:1 line represents perfect quantile agreement. MAQE denotes mean absolute quantile error, while Q₉₅ and Q₉₉ biases represent model minus ERA5-Land quantiles.

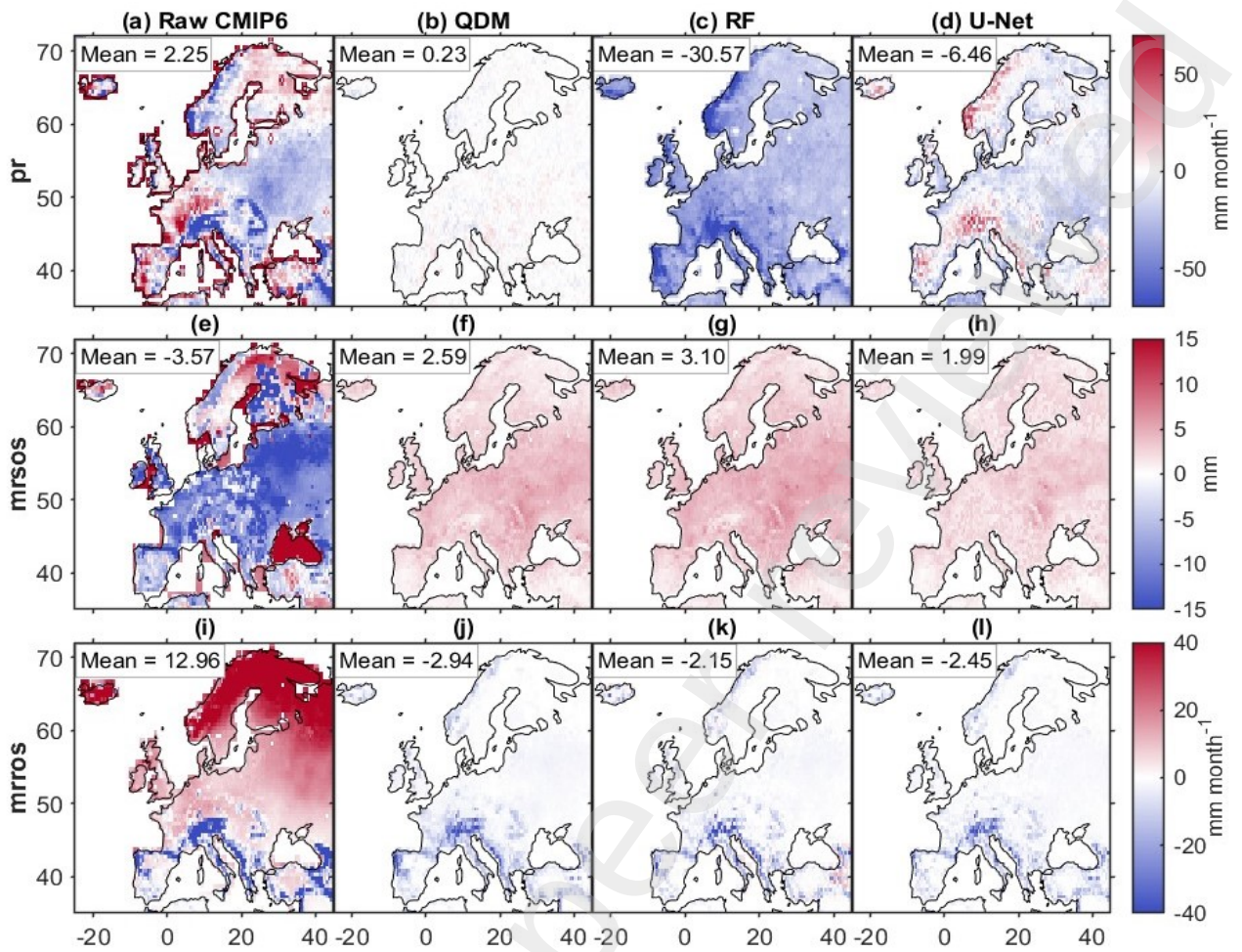


Figure 9. Spatial bias in selected extreme percentiles relative to ERA5-Land during the independent testing period. Maps show model percentile minus ERA5-Land percentile for raw, QDM-, RF- and U-Net-corrected simulations. Precipitation and surface runoff are evaluated using P95, while surface soil moisture is evaluated using P5. Positive values indicate overestimation and negative values indicate underestimation.

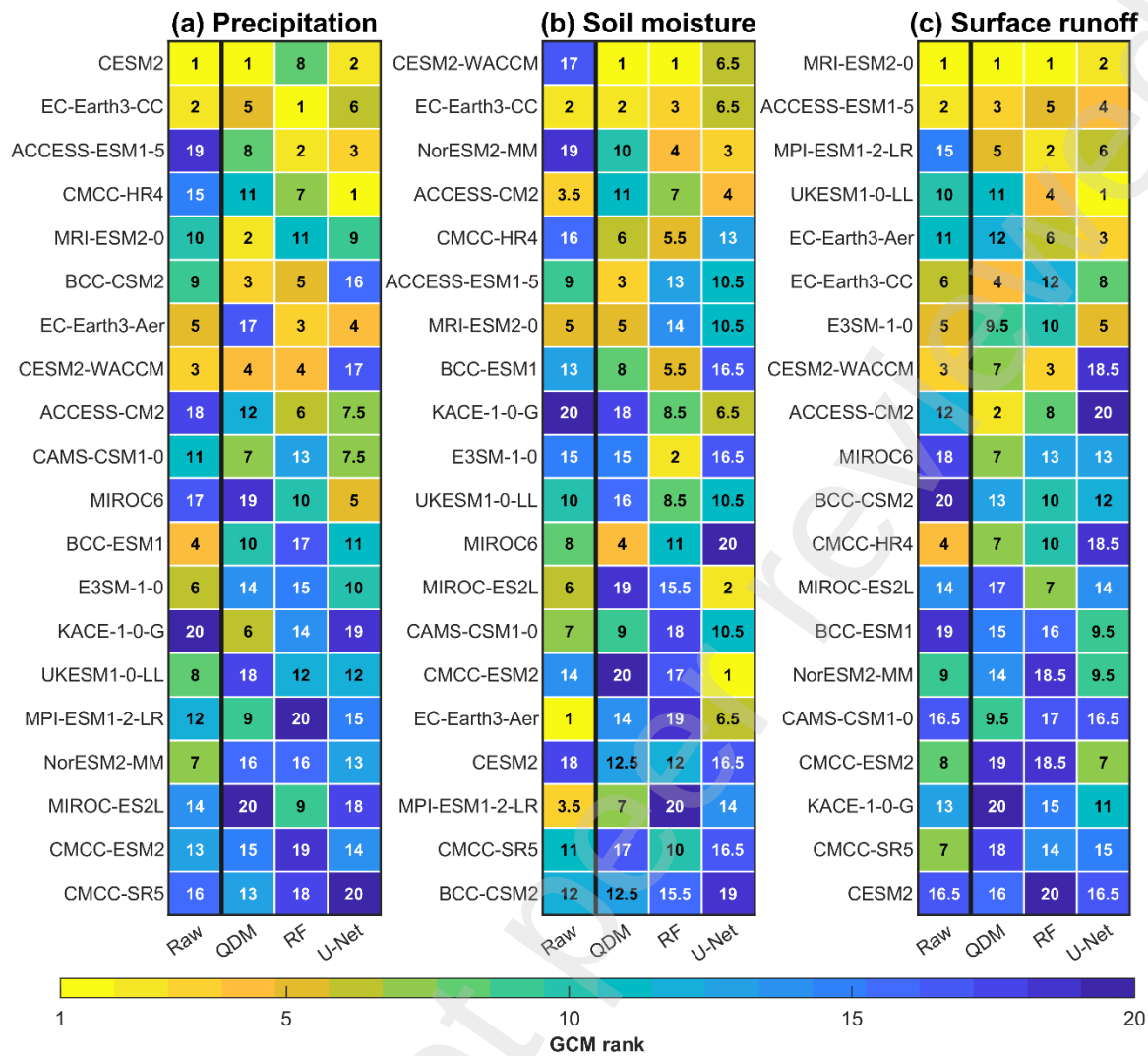


Figure 10. TOPSIS-based ranking of the 20 CMIP6 models for precipitation, surface soil moisture and surface runoff under raw, QDM-, RF- and U-Net-corrected conditions. Rankings combine KGE as a benefit criterion and RMSE as a cost criterion using weights of 0.60 and 0.40, respectively. Lower rank numbers indicate stronger combined performance.

Comparative Performance of Statistical, Machine-Learning and Deep-Learning Bias-Correction Frameworks for CMIP6 Precipitation, Soil Moisture and Runoff across Europe

Fatimatuj Sonny^{a,*}, Sogol Moradian^b, Mohammad Javad Alizadeh^a, Agnieszka Indiana Olbert^a

^a Department of Civil Engineering, College of Science and Engineering, University of Galway, Galway, Ireland.

^b Department of Civil Engineering and Construction, Atlantic Technological University, Sligo, Ireland.

Abstract

Bias correction is widely used to improve the suitability of climate-model simulations for regional hydro-climatic applications; but climate-model bias is multidimensional and can affect magnitude, variability, temporal agreement, spatial patterns and distributional characteristics. These biases also differ among precipitation, soil moisture and runoff because the variables have contrasting statistical behaviour, temporal persistence and spatial structure. As a result, the relative performance of statistical, machine-learning and deep-learning correction methods cannot be fully assessed using a single variable or evaluation metric, yet consistent cross-variable comparisons remain limited. This study compares Quantile Delta Mapping (QDM), Random Forest (RF) and variable-specific U-Net-based frameworks for precipitation, surface soil moisture and surface runoff from 20 climate models from the Coupled Model Intercomparison Project Phase 6 (CMIP6) across Europe, using ERA5-Land as a common reference. Performance was assessed under consistent spatial, temporal and independent testing conditions using domain-level statistics, spatial maps of changes in grid-cell temporal correlation and distributional diagnostics. The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) was also used to examine changes in individual model rankings. The U-Net-based configurations produced the strongest overall multi-metric performance, increasing correlation from 0.170 to 0.689 for precipitation, 0.464 to 0.750 for soil moisture and 0.164 to 0.706 for runoff. They also achieved the highest Nash–Sutcliffe efficiency and Kling–Gupta efficiency values and the largest spatial mean increases in grid-cell correlation of 0.483, 0.274 and 0.537, respectively. However, QDM showed stronger marginal-distribution and quantile agreement, particularly for precipitation and soil moisture, while RF reduced several magnitude-based errors but gave less consistent improvements in correlation and efficiency. Bias correction also narrowed historical inter-model dispersion and altered TOPSIS rankings, particularly for precipitation and soil moisture, whereas runoff rankings were more persistent. Overall, bias-correction performance was variable- and diagnostic-dependent, indicating that method selection should reflect the intended application and that climate-model ranking should be conducted after the intended correction is applied.

Keywords: Climate change; Climate model bias correction; Quantile Delta Mapping; Random Forest; U-Net; Multi-model ensemble.

1. Introduction

Global climate models are fundamental tools for characterising climate variability and assessing hydro-climatic responses to a changing climate, with the Coupled Model Intercomparison Project Phase 6 (CMIP6) providing a coordinated multi-model framework for evaluating climate processes and responses to external forcing^{1,2}. Nevertheless, CMIP6 simulations retain systematic discrepancies relative to observational and reanalysis-based reference datasets, particularly at regional scales and for impact-relevant hydro-climatic variables³⁻⁵. Biases in the magnitude, variability and distribution of precipitation, soil moisture and runoff can affect assessments of water availability, drought and land-surface hydrological response, while their characteristics vary among models, regions and variables⁴⁻⁶. Bias correction is therefore often required to improve the suitability of climate-model output for regional applications. However, improvement in marginal distributions or mean error does not

necessarily imply corresponding improvement in temporal agreement, spatial structure or other properties of the simulated fields^{4,7}.

Climate-model bias can be broadly defined as a systematic difference between a simulated climate statistic and the corresponding characteristic of a reference dataset⁴. Such biases arise from interacting sources including finite spatial resolution, uncertain parameterisations, imperfect representation of topography and land-surface heterogeneity, as well as limitations in resolving sub-grid processes such as convection, cloud formation, snow processes and land–atmosphere exchanges^{3,7,8}. Their expression is variable dependent: precipitation biases are strongly influenced by atmospheric and orographic processes, whereas soil-moisture and runoff biases are additionally affected by land-surface parameterisation, soil-layer structure, storage and water redistribution^{9,10}. Consequently, biases may affect the mean state, variability, seasonal cycle, distributional shape and distribution tails, with substantial differences among models and regions. Bias correction seeks to reduce these model–reference differences by deriving relationships from a historical calibration or training period and applying them to model output^{3,4}. However, it modifies selected statistical characteristics rather than correcting the underlying structural deficiencies of the climate model, so different approaches may improve different aspects of the simulated fields^{7,11}.

Bias correction is challenging across the terrestrial water cycle because precipitation, soil moisture and runoff differ markedly in their statistical structure, temporal memory and spatial controls. Precipitation is intermittent, highly variable and often strongly skewed. Its variability is influenced by atmospheric circulation, convection, topography and seasonality^{9,11}. Surface soil moisture is a bounded storage variable with stronger temporal persistence, integrating antecedent precipitation, evapotranspiration, soil properties, vegetation and land-surface conditions^{10,12}. Runoff also has a skewed distribution and responds nonlinearly to precipitation, antecedent wetness, snowmelt and land-surface storage^{13,14}. These differences affect the magnitude, persistence and spatial pattern of model biases. They therefore need to be considered when developing and evaluating bias-correction methods for different hydro-climatic variables.

Statistical bias-correction methods range from simple mean and variance adjustments to distribution-based approaches such as quantile mapping^{4,6}. Quantile-based methods are widely used because they adjust biases across the simulated distribution rather than correcting only the mean state. Quantile Delta Mapping (QDM) further preserves quantile-specific modelled changes while correcting systematic differences between model and reference distributions¹¹. This makes it suitable for hydro-climatic variables with skewed and spatially heterogeneous distributions. However, QDM is generally applied independently at individual grid cells and primarily targets marginal distributions without explicitly representing nonlinear spatial relationships among neighbouring locations. These limitations have encouraged the development of more flexible machine-learning and spatially informed correction approaches.

Machine-learning methods can learn nonlinear relationships between climate-model simulations and reference data without requiring a predefined statistical mapping. Random Forest (RF), for example, uses ensembles of decision trees to represent nonlinear interactions among predictors and has been increasingly applied to climate-model bias correction¹⁵. Recent applications have demonstrated its potential, either alone or combined with quantile-based correction, to reduce systematic precipitation biases in CMIP6 simulations. However, tree-based methods do not inherently exploit the spatial organisation of complete gridded fields unless neighbouring information is explicitly supplied as predictors. Deep-learning methods, particularly convolutional neural networks, can instead learn spatially organised features directly from gridded climate data and across multiple spatial scales¹⁶. U-Net is well suited to such field-to-field correction because its encoder–decoder architecture captures broader spatial features while skip connections retain finer-scale information during reconstruction¹⁷.

Recent applications have demonstrated the potential of U-Net and related convolutional frameworks to improve the statistical performance of bias-corrected precipitation fields^{16,18}.

Despite substantial advances in climate-model bias correction, consistent comparison across methods, hydro-climatic variables and evaluation frameworks remains limited. Existing studies differ widely in target variables, climate models, reference datasets, spatial scales and performance criteria, making direct comparison of statistical, machine-learning and deep-learning approaches difficult^{4,7,19}. Recent machine-learning and convolutional applications have also focused largely on precipitation and other atmospheric variables, with comparatively limited assessment of precipitation, surface soil moisture and surface runoff within the same CMIP6 ensemble, reference dataset and independent testing framework^{15,16,18,20}. Performance is also often summarised using a limited set of statistics, although different diagnostics can lead to contrasting assessments of correction performance^{7,11,19}. In addition, individual CMIP6 models may rank differently depending on the evaluation criteria, while the extent to which bias correction alters these rankings remains insufficiently examined. Multi-criteria approaches such as the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) provide a structured means of combining complementary performance measures for model selection²¹⁻²³. These gaps motivate a common multi-variable, multi-diagnostic and multi-criteria evaluation framework.

To address these gaps, this study establishes a common framework for evaluating statistical, machine-learning and deep-learning bias-correction approaches for CMIP6 precipitation, surface soil moisture and surface runoff across Europe. Twenty CMIP6 models are evaluated against ERA5-Land using QDM, RF and variable-specific U-Net-based configurations under consistent spatial, temporal and independent testing conditions. The main contribution is a unified comparison of these method families within a common multi-variable framework, extending assessment beyond conventional error statistics to include correlation, hydrological efficiency, spatial skill, distributional agreement, monthly distribution tails and inter-model dispersion. The framework also examines how bias correction modifies the relative ranking of individual CMIP6 models. Specifically, the objectives are to: (i) quantify and compare the performance of QDM, RF and U-Net-based correction frameworks for precipitation, surface soil moisture and surface runoff; (ii) characterise their effects on the statistical, spatial, distributional and inter-model properties of the corrected fields; and (iii) determine how different bias-correction approaches influence the subsequent ranking and selection of individual CMIP6 models. This integrated framework provides a basis for identifying the capabilities, trade-offs and limitations of alternative bias-correction approaches for regional hydro-climatic applications.

2. Materials and methodology

2.1 Materials

2.1.1 Study area

The study domain covers Europe, extending from 35°N-72°N and from 25°W- 45°E. Europe provides a suitable region for comparing bias-correction methods because it contains strong climatic, topographic and hydrological gradients within a single continental domain. Western and north-western Europe are mainly influenced by humid maritime conditions associated with the North Atlantic, whereas eastern Europe shows stronger continental behaviour and greater seasonal contrasts. Northern Europe includes colder and snow-influenced environments, while southern Europe and the Mediterranean are characterised by pronounced precipitation seasonality and relatively dry summers^{2,24}.

European hydro-climatic conditions are further modified by major mountain systems, including the Alps, Pyrenees, Carpathians and Scandinavian Mountains. Elevation and orographic effects influence the spatial distribution of precipitation, snow accumulation and melt, soil-water storage and runoff generation. The combined effects of latitude, distance from the Atlantic Ocean, continentality, Mediterranean circulation and topography therefore produce considerable regional variability in

precipitation and land-surface hydrological behaviour across relatively short distances^{24,25}. Major mountain systems, including the Alps, Pyrenees, Carpathians and Scandinavian Mountains, further modify precipitation, snow accumulation, soil-water storage and runoff generation through elevation and orographic effects. These combined climatic and topographic controls produce substantial spatial variability in precipitation, surface soil moisture and surface runoff across Europe²⁴.

European hydrological responses vary considerably among regions. Europe-wide studies have identified distinct hydro-climatic and flood-seasonality regions, including Atlantic, Continental, Scandinavian, Alpine and Mediterranean regimes, confirming the strong regional diversity of hydrological response across the continent²⁵. These regional contrasts reflect differences in precipitation seasonality, snow accumulation and melt, antecedent soil-moisture conditions, topography and catchment characteristics. This diversity makes Europe an appropriate test area for evaluating whether statistical, machine-learning and deep-learning bias-correction methods perform consistently across contrasting climatic and land-surface settings.

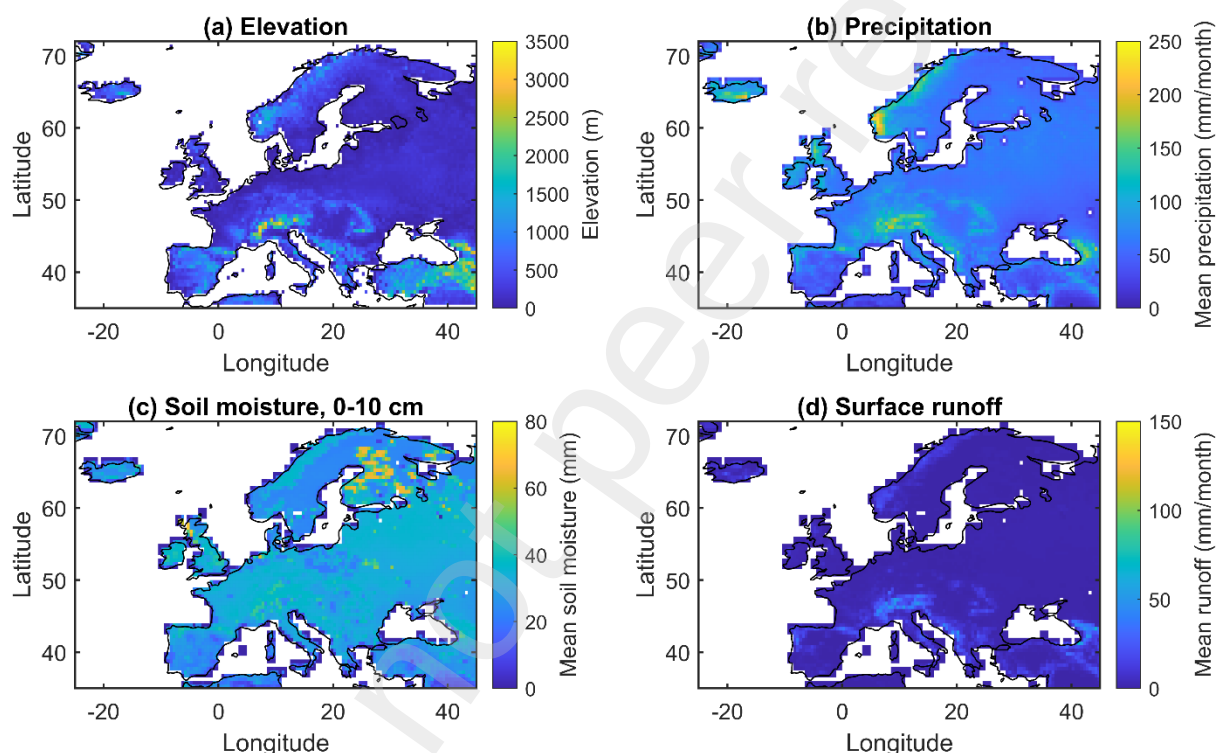


Figure 1. European study domain and spatial distribution of elevation and ERA5-Land reference precipitation, surface soil moisture and surface runoff during 1980–2014.

2.1.2 Datasets

The study combines reanalysis and climate-model datasets to characterise key land-surface hydro-climatic conditions across Europe. Three variables were considered: precipitation, surface soil moisture and surface runoff, representing atmospheric water input, near-surface water storage and land-surface outflow, respectively. ERA5-Land was used as the reference reanalysis datasets for evaluating the CMIP6 simulations over the common period January 1980–December 2014, corresponding to 420 monthly time steps. ERA5-Land is produced by the European Centre for Medium-Range Weather Forecasts and provides globally consistent land-surface variables derived from the ERA5 atmospheric reanalysis at an approximate native spatial resolution of 0.1° ^{26,27}. It was selected as it provides spatially continuous and physically consistent fields for precipitation, soil moisture and runoff within a single framework. However, ERA5-Land was treated as a reference reanalysis dataset rather than as an error-free observational dataset.

The ERA5-Land variables used were total precipitation (tp), surface runoff (sro) and volumetric soil water in the first two soil layers (swvl1 and swvl2). Precipitation and surface runoff were represented as monthly totals in mm/month. Surface soil moisture was defined as the equivalent water depth in the upper 10 cm of the soil column. This was derived by combining the complete 0–7 cm ERA5-Land layer with the upper 3 cm of the 7–28 cm layer:

$$SM_{0-10} = 70 \text{ swvl1} + 30 \text{ swvl2} \quad (1)$$

where swvl1 and swvl2 are expressed in $m^3.m^{-3}$, and the resulting soil-water depth is expressed in millimetres.

Climate-model simulations were obtained from the CMIP6 historical experiment¹. The variables used were precipitation (pr), surface soil moisture (mrsos) and surface runoff (mrros). A total of 20 global climate models were retained based on the continuous availability of all three variables over 1980–2014. The same model set was used for the three variables to support a consistent cross-variable comparison. The selected models represent different modelling centres, atmospheric configurations, land-surface schemes and native spatial resolutions. CMIP6 mrsos was treated as the corresponding surface soil-moisture variable, although differences in effective soil-layer representation among models may remain. Model names, institutions and native resolutions are provided in Table 1.

Table 1. CMIP6 models used in this study, including modelling institution, full model name and native spatial resolution.

CMIP6 Model	Institution	Model Abbreviation	Resolution	Reference
ACCESS-CM2	CSIRO-ARCCSS	Australian Community Climate and Earth System Simulator Climate Model Version 2	1.25°×1.875°	Bi et al. (2020) ²⁸
ACCESS-ESM1-5	CSIRO	Australian Community Climate and Earth System Simulator Earth System Model Version 1.5	1.25°×1.875°	Ziehn et al. (2020) ²⁹
BCC-CSM2-MR	BCC	Beijing Climate Centre Climate System Model version 2, Medium Resolution	1.125°×1.125°	Wu et al. (2019) ³⁰
BCC-ESM1	BCC	Beijing Climate Centre Earth System Model version 1	2.8° × 2.8°	Wu et al. (2020) ³¹
CAMS-CSM1-0	CAMS	Chinese Academy of Meteorological Sciences Climate System Model version 1.0	1.125°×1.125°	Rong et al. (2018) ³²
CESM2	NCAR	Community Earth System Model version 2	0.9° × 1.25°	Danabasoglu et al. (2020) ³³
CESM2-WACCM	NCAR	Community Earth System Model version 2 with Whole Atmosphere Community Climate Model	0.9° × 1.25°	Danabasoglu et al. (2020) ³³
CMCC-CM2-HR4	CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici Climate Model 2, High Resolution 4	0.94° × 1.25°	Scoccimarro et al. (2022) ³⁴
CMCC-CM2-SR5	CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici Climate Model 2, Standard Resolution 5	0.94° × 1.25°	Cherchi et al. (2019) ³⁵
CMCC-ESM2	CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici Earth System Model 2	0.94° × 1.25°	Lovato et al. (2022) ³⁶
E3SM-1-0	E3SM-Project	Energy Exascale Earth System Model version 1.0	1.0° × 1.0°	Golaz et al. (2019) ³⁷
EC-Earth3-AerChem	EC-Earth Consortium	EC-Earth3 with interactive aerosols and atm chemistry	0.7° × 0.7°	van Noije et al. (2021) ³⁸
EC-Earth3-CC	EC-Earth Consortium	EC-Earth3 carbon cycle configuration	0.7° × 0.7°	Döscher et al. (2022) ³⁹
KACE-1-0-G	NIMS-KMA	Korean Advanced Community Earth System Model version 1.0, GLOMAP	1.25° × 1.875°	Lee et al. (2020) ⁴⁰
MIROC6	MIROC	Model for Interdisciplinary Research on Climate version 6	1.4° × 1.4°	Tatebe et al. (2019) ⁴¹
MIROC-ES2L	MIROC	MIROC Earth System Model version 2 for Long-term simulations	2.8° × 2.8°	Hajima et al. (2020) ⁴²
MPI-ESM1-2-LR	MPI-M, AWI, DKRZ, DWD	Max Planck Institute Earth System Model version 1.2, Low Resolution	1.87°×1.87°	Mauritsen et al. (2019) ⁴³
MRI-ESM2-0	MRI	Meteorological Research Institute Earth System Model version 2.0	1.12° × 1.12°	Yukimoto et al. (2019) ⁴⁴
NorESM2-MM	NCC	Norwegian Earth System Model version 2, Low atmosphere-medium ocean resolution	1.9° × 2.5°	Seland et al. (2020) ⁴⁵
UKESM1-0-LL	MOHC, NERC, NIMS-KMA, NIWA	UK Earth System Model version 1.0, Low Resolution	1.25° × 1.875°	Sellar et al. (2019) ⁴⁶

2.2 Preprocessing

All ERA5-Land and CMIP6 datasets were processed using a common workflow before the bias-correction experiments. Monthly records were first aligned using common year-month indices to account for differences in calendar conventions among CMIP6 models. Longitude conventions, latitude orientation and temporal dimensions were harmonised, after which the fields were cropped to the European study domain. Only simulations with complete temporal coverage over the analysis period were retained. Where multiple valid historical variants were available for a given model and variable, each variant was processed separately, and the aligned fields were averaged to obtain one representative simulation for that model. This reduced sensitivity to individual ensemble members while maintaining a consistent model archive.

The datasets were first harmonised in terms of temporal indexing, coordinate structure and variable units, following the definitions provided in Section 2.1. All ERA5-Land and CMIP6 fields were then interpolated onto a common $0.5^\circ \times 0.5^\circ$ latitude-longitude grid using bilinear interpolation. The same predefined target grid was used for all variables, climate models and correction methods, producing spatially paired CMIP6 and ERA5-Land fields at each grid cell. Bilinear interpolation was selected because it provides continuous gridded fields suitable for spatial machine-learning and deep-learning applications. However, it may smooth local precipitation maxima, runoff extremes and sharp soil-moisture gradients, particularly over mountainous, coastal and hydrologically heterogeneous regions. This effect was considered when interpreting local and extreme-percentile performance.

A common land mask derived from ERA5-Land was applied after spatial alignment to ensure that all datasets shared the same evaluation domain. Only terrestrial grid cells with valid reference values were retained throughout the analysis. Ocean grid cells were preserved as NaN in the processed datasets and excluded from subsequent correction and performance calculations. This prevented artificial values over non-land areas and ensured that all methods were evaluated using the same set of valid European land grid cells. The processed monthly datasets were divided chronologically into model-development and independent testing periods to preserve the temporal structure of the records and prevent information from later years entering model calibration. The first 80% of the monthly sequence was assigned to model development, while the final 20% was retained for independent testing. The same temporal division, target grid, land mask, variable units and evaluation domain were used for all bias-correction methods. Further details on the use of the development data for training and validation, together with method-specific handling of missing values and model inputs, are provided in Section 2.4.

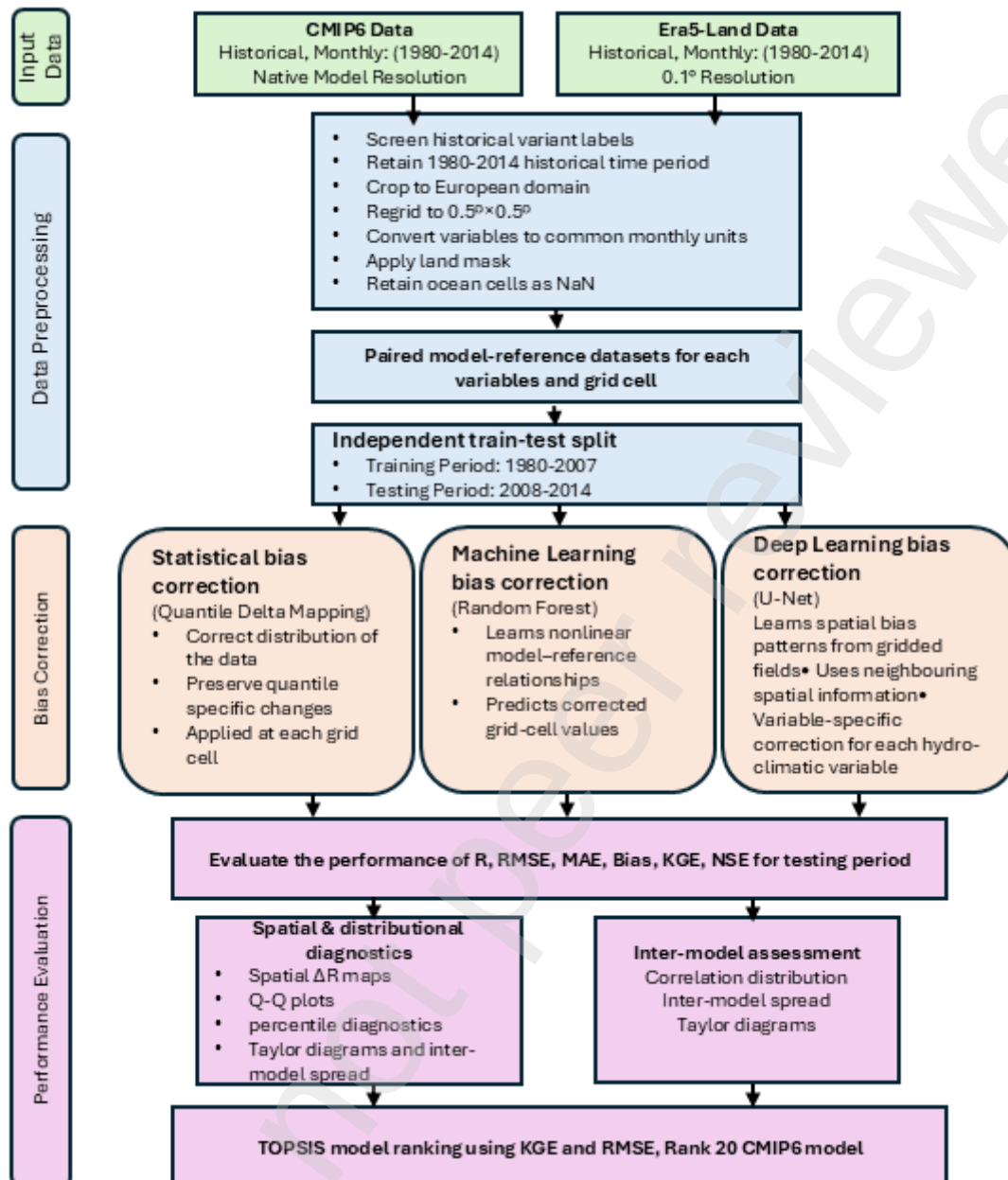


Figure 2. Overall workflow for preprocessing, bias correction and performance evaluation of the ERA5-Land and CMIP6 datasets.

2.3 Bias-correction framework

A common evaluation framework was developed to compare the final QDM, RF and U-Net-based correction configurations under consistent spatial, temporal and evaluation conditions. For each CMIP6 model and hydro-climatic variable, the pre-processed historical simulation was used as the biased input, while the corresponding ERA5-Land field served as the reference target. Correction was conducted separately for precipitation, surface soil moisture and surface runoff because these variables differ in their distributions, temporal persistence, spatial organisation and bias structures.

The raw CMIP6 simulations were first evaluated against ERA5-Land to establish the baseline against which improvements from the correction methods were measured. Each method was calibrated or trained using the model-development period and then applied to the independent testing period. QDM provided a grid-cell-based statistical correction of the empirical distributions, RF represented nonlinear relationships between model predictors and ERA5-Land values, and U-Net was used to learn spatially

organised field-to-field correction patterns. The full methodological formulation and model settings are described in Section 2.4.

Each CMIP6 model was corrected independently before evaluation. Individual corrected outputs were retained to examine inter-model differences, while multi-model ensemble means were calculated only after correction for the overall comparison with ERA5-Land. All methods used the same CMIP6 model set, reference dataset, target grid, land mask, model-development and testing periods, and evaluation metrics. However, the final correction configurations included method-specific predictor, spatial and temporal structures. The comparison therefore assesses the performance of the complete correction configurations rather than isolating the effect of algorithm type alone.

2.4 Bias-correction methods

2.4.1 Statistical bias correction

QDM was used as the statistical bias-correction method in this study. QDM is a distribution-based approach that adjusts systematic differences between climate-model and reference distributions while retaining the quantile-dependent characteristics of the model simulation¹¹. It was selected as an established statistical benchmark because precipitation, surface soil moisture and surface runoff commonly exhibit skewed distributions, nonlinear biases and strong spatial variability that cannot be adequately represented through adjustment of the mean alone^{4,6,11}.

The correction was performed independently for each CMIP6 model, hydro-climatic variable, calendar month and valid land grid cell. Separate monthly mappings were developed to account for seasonal differences in the statistical distributions and bias structures of the three variables. The empirical CMIP6 and ERA5-Land distributions for each calendar month were estimated using data from the calibration period. For each CMIP6 value x_m , its corresponding quantile position was determined as:

$$q = F_m(x_m) \quad (2)$$

where F_m represents the empirical cumulative distribution function of the CMIP6 calibration series and q is the cumulative probability associated with the value being corrected. The corresponding model and reference quantiles based on Equation 2 were then used internally to calculate the QDM correction while retaining the quantile-specific model departure.

Linear interpolation was used to estimate probabilities between the empirical values, while values outside the calibration range were assigned to the nearest boundary quantile. Grid cells with insufficient valid paired data were excluded, and negative corrected values were set to zero. The monthly QDM relationships were calibrated using 1980–2007 and evaluated independently during 2008–2014.

2.4.2 Machine-learning bias correction

RF was used as the machine-learning bias-correction method. RF is an ensemble regression algorithm that combines predictions from multiple decision trees, thereby allowing nonlinear relationships to be represented while reducing sensitivity to any single tree⁴⁷. It was selected because biases in precipitation, surface soil moisture and surface runoff may vary according to simulated magnitude, season and geographical location^{15,20}.

The RF correction was implemented separately for each CMIP6 model and hydro-climatic variable. The temporal configuration was adapted to the characteristics of each variable to account for differences in seasonality and bias structure. For each model configuration, the predictor variables consisted of the CMIP6 value at each valid land grid cell together with the corresponding latitude and longitude, while the paired ERA5-Land value was used as the prediction target. Including latitude and longitude allowed RF to represent broad geographical variations in model bias across Europe.

The models were trained using records from 1980–2007. Before fitting, the monthly gridded fields were converted into individual samples, and only rows with finite predictor and target values were retained.

This procedure excluded ocean and missing grid cells from training. The final RF configuration used 300 regression trees, a maximum tree depth of 20, a minimum of two samples per terminal leaf and square-root feature selection at each split. A random seed of 42 was specified to ensure reproducibility, while parallel processing was used to reduce computational time.

After training, each monthly RF model was applied to all corresponding monthly CMIP6 fields to generate the corrected spatial outputs. Predictions were produced only for grid cells with valid input values, while invalid cells remained NaN. Negative corrected values were subsequently set to zero to maintain physically meaningful values for precipitation and surface runoff. Although corrected outputs were generated for the complete historical period, performance statistics were calculated independently for 2008–2014 using the metrics described in Section 2.5.

2.4.3 Deep-learning bias correction

U-Net was adopted as the deep-learning bias-correction method because its convolutional encoder–decoder structure is well suited to spatial field-to-field prediction¹⁷. Climate-model errors are often spatially organised across neighbouring grid cells rather than occurring independently at individual locations. Convolution-based architectures can therefore improve the representation of regional bias patterns and spatial dependence that are not explicitly considered by conventional grid-cell-based correction methods^{16,48}. Within U-Net, the encoder progressively extracts local and broader spatial features, while the decoder reconstructs the corrected field at the original grid resolution. Skip connections transfer high-resolution information directly between corresponding encoder and decoder levels, helping preserve fine-scale gradients and spatial structures during reconstruction. U-Net was therefore selected to complement QDM and RF by explicitly representing the spatial organisation of precipitation, surface soil moisture and surface runoff and by learning nonlinear relationships between CMIP6 simulations and ERA5-Land reference fields.

The U-Net backbone consisted of three encoder levels with 32, 64 and 128 filters, followed by a bottleneck with 256 filters and three decoder levels with 128, 64 and 32 filters. Each convolutional block contained two 3×3 convolutional layers with same padding, ReLU activation and batch normalisation. Batch normalisation was used to stabilise the distribution of intermediate feature activations and improve training efficiency. Max-pooling was applied between encoder levels to reduce the spatial dimensions and allow the network to extract progressively broader spatial features. In the decoder, transposed convolutions restored the spatial resolution, while skip connections transferred high-resolution features from each encoder level to the corresponding decoder level. This allowed the network to combine broad regional information with local gradients and fine-scale spatial detail during reconstruction. Dropout was applied in the deeper part of the network to reduce overfitting and improve generalisation to unseen testing data. The final layer used a 1×1 convolution with linear activation to generate one continuous corrected output field at the original grid resolution. The same U-Net backbone was retained for precipitation, surface soil moisture and surface runoff, while the input–target formulation, temporal structure and loss function were adapted to the specific bias behaviour and temporal characteristics of each variable. The complete network architecture is shown in Figure 3.

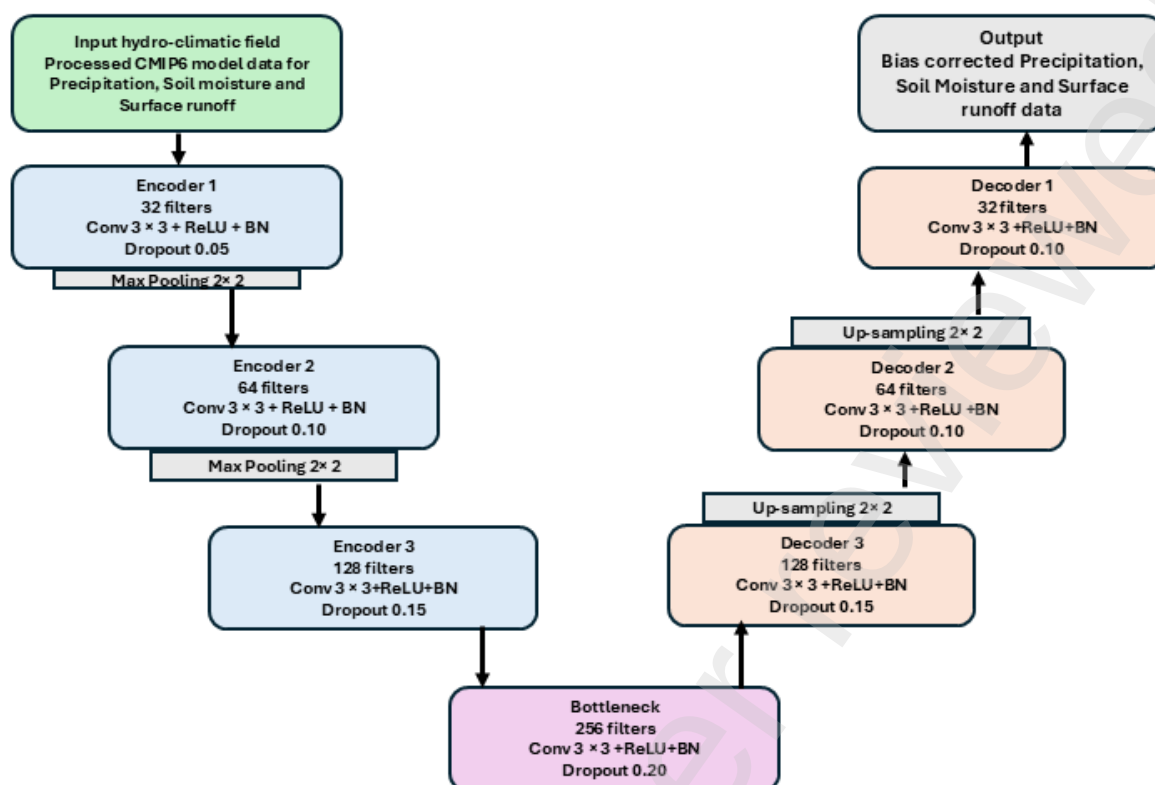


Figure 3. U-Net encoder–decoder architecture used for spatial bias correction of CMIP6 hydro-climatic fields.

The U-Net experiments followed a common season-specific and chronological framework, while the input–target formulation, temporal context and network depth were adapted to the characteristics of each hydro-climatic variable. One monthly spatial field was treated as one sample, and separate networks were trained for DJF, MAM, JJA and SON. Pre-2008 records were used to estimate the network weights, while records from January 2008 to December 2014 were used for testing period on which performance assessment has been evaluated. Validation loss was monitored to adjust the learning rate, determine the early-stopping epoch and restore the model weights associated with the lowest validation loss. All models used Adam optimization, although the learning rate, batch size, maximum number of epochs, early-stopping patience, dropout structure, loss function and number of trainable parameters varied among the three variable-specific configurations. The exact training and validation periods, seasonal sample numbers and complete optimization settings are reported in Supplementary Table S1.

Before identifying the final models, several correction strategies were examined, including direct field prediction, anomaly-based correction, residual learning, seasonal and non-seasonal training, lagged and non-lagged inputs, auxiliary static predictors, alternative dropout settings and different loss functions. The final configuration for each variable was selected according to validation performance, training stability and the physical consistency of the corrected spatial fields. Ocean pixels were assigned a numerical fill value before being passed to the network, while a masked loss function ensured that only valid terrestrial grid cells contributed to model optimisation. Ocean cells were restored to NaN after prediction and excluded from the evaluation.

For precipitation, the selected approach combined QDM with seasonal U-Net correction. The QDM-corrected precipitation field was used as the main dynamic input, while elevation was included as a static auxiliary predictor to help represent topographically influenced precipitation patterns. The network directly predicted the corresponding ERA5-Land precipitation field. Separate models were developed for DJF, MAM, JJA and SON to account for pronounced seasonal differences in European precipitation processes and model biases. The precipitation configuration was optimized using a

combined loss comprising MAE, correlation loss and variance loss. This formulation was retained because it improved precipitation magnitude, temporal correlation and variability simultaneously.

For surface soil moisture, the selected configuration used a seasonal lagged-anomaly U-Net to account for the relatively strong temporal persistence of near-surface soil moisture. Monthly climatologies were first calculated separately for CMIP6 and ERA5-Land using the model-development period, and the corresponding monthly anomalies were then derived. Three consecutive CMIP6 anomaly fields at t , $t-1$ and $t-2$ were used as dynamic input channels, together with sine and cosine representations of calendar month. The prediction target was the corresponding ERA5-Land soil-moisture anomaly at month t . Separate U-Net models were trained for DJF, MAM, JJA and SON using MAE loss. After prediction, the anomalies were transformed back to the original units, and the ERA5-Land monthly climatology was added to obtain the final corrected soil-moisture fields. This formulation allowed the network to combine spatial information from neighbouring grid cells with short-term soil-moisture persistence and seasonal variability.

For surface runoff, the selected configuration combined QDM with a seasonal lagged residual U-Net. This formulation was adopted because runoff is strongly influenced by antecedent wetness, land-surface and catchment storage, and delayed hydrological response. After QDM correction, monthly runoff anomalies were calculated, and seven consecutive QDM-corrected anomaly fields from $t-6$ to t were supplied to the network. The prediction target was the residual anomaly between ERA5-Land and QDM-corrected runoff at month t . The predicted residual was then added to the QDM-corrected runoff field to obtain the final corrected output. This lagged residual design provided a more stable representation of runoff variability than direct correction or non-lagged anomaly formulations. MAE was used as the final loss function because it produced the most robust residual-learning performance. The final variable-specific configurations are summarised in Table 2.

Table 2. U-Net bias-correction configurations for precipitation, surface soil moisture and surface runoff.

Variable	Correction design	Model inputs	Prediction target	Temporal structure	Loss function
Precipitation	QDM followed by direct U-Net correction	QDM-corrected precipitation and elevation	ERA5-Land precipitation	Separate seasonal models for DJF, MAM, JJA and SON	MAE + correlation loss + variance loss
Surface soil moisture	Seasonal lagged-anomaly U-Net	CMIP6 soil moisture anomalies	ERA5-Land soil moisture anomaly	Separate DJF, MAM, JJA and SON models using three lagged anomaly fields	MAE
Surface runoff	QDM followed by lagged residual U-Net correction	QDM-corrected runoff anomalies from ($t-6$) to (t)	Residual anomaly between ERA5-Land and QDM-corrected runoff at month (t)	Separate seasonal models using seven consecutive monthly inputs	MAE

2.5 Performance evaluation

Performance was evaluated by comparing the raw and bias-corrected CMIP6 fields with ERA5-Land during January 2008–December 2014, comprising 84 monthly observations. Each CMIP6 model was corrected independently before the corrected fields were averaged to produce a multi-model ensemble mean for QDM, RF and U-Net. The raw simulations were averaged in the same way to produce the uncorrected ensemble baseline. All methods were evaluated using the same temporal period, spatial grid, land mask and paired-data requirement.

Six complementary metrics were calculated: the Pearson correlation coefficient (R), root mean square error (RMSE), mean absolute error (MAE), mean bias, Nash–Sutcliffe efficiency (NSE) and Kling–Gupta efficiency (KGE). R measured the temporal association between the simulated and ERA5-Land monthly series. RMSE and MAE quantified error magnitude, with RMSE assigning greater influence to large errors. Bias was calculated as the simulated value minus the corresponding ERA5-Land value,

such that positive and negative values indicated overestimation and underestimation, respectively. NSE assessed predictive performance relative to the mean of the ERA5-Land series, while KGE combined correlation, relative variability and the ratio between the simulated and reference means.

The metrics were calculated separately from the monthly time series at each valid European land grid cell and were subsequently averaged across the spatial domain. Thus, the values in Table 3 represent spatial means of grid-cell temporal metrics, rather than metrics calculated from a flattened space–time array or arithmetic means of the individual-model statistics. Grid cells containing fewer than 10 paired monthly values were excluded. For surface runoff, cells with an ERA5-Land standard deviation below 0.1 mm month or an absolute mean below 0.1 mm month were also excluded to limit instability in NSE and KGE under very low-flow and low-variability conditions. The resulting reference-based reliability mask was applied consistently to the raw, QDM-, RF- and U-Net-corrected runoff fields. Ocean and missing grid cells were excluded from all calculations. Higher R , NSE and KGE values indicated improved performance, while lower RMSE, MAE and absolute bias indicated reduced error. Inter-model distributions, uncertainty envelopes, Q–Q plots, selected-percentile biases and Taylor diagrams were also used to assess model consistency, distributional agreement, variability and overall performance⁴⁹.

2.6 Multi-criteria CMIP6 model ranking

The TOPSIS method was applied to rank the individual CMIP6 models according to their combined statistical performance^{21,50}. The analysis was performed separately for precipitation, surface soil moisture and surface runoff and independently for the raw, QDM-, RF- and U-Net-corrected simulations. Each CMIP6 model was treated as an alternative, while KGE and RMSE were selected as complementary decision criteria. The joint use of KGE and RMSE or normalized RMSE is consistent with recent multi-criteria evaluations and rankings of CMIP6 models^{51–53}. KGE was selected because it combines correlation, relative variability and mean bias within a single measure, whereas RMSE provides a direct measure of overall error magnitude and gives greater influence to large deviations. Restricting the analysis to these two criteria reduced redundancy with the other calculated metrics, as correlation and bias are represented within KGE, while NSE and RMSE both depend on squared errors. KGE was specified as a benefit criterion because higher values indicate stronger overall agreement with ERA5-Land, whereas RMSE was specified as a cost criterion because lower values indicate smaller model–reference errors.

For each variable and correction method, a decision matrix was constructed from the KGE and RMSE values of all models. As the criteria have different units and numerical ranges, vector normalisation was applied:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}} \quad (3)$$

where x_{ij} is the value of criterion j for model i . The normalised values were then weighted as:

$$v_{ij} = w_j r_{ij} \quad (4)$$

Weights of 0.60 and 0.40 were assigned to KGE and RMSE, respectively, giving greater emphasis to overall hydrological performance while retaining sensitivity to error magnitude. Positive and negative ideal solutions were defined from the best and worst weighted criterion values. The relative closeness coefficient was calculated as:

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (5)$$

where D_i^+ and D_i^- are the distances from the positive and negative ideal solutions. Models with C_i values closer to one were assigned higher ranks.

3. Results

This section presents the performance of QDM, RF and U-Net in correcting CMIP6 precipitation, surface soil moisture and surface runoff across Europe. The raw CMIP6 simulations were used as the uncorrected baseline, and all corrected outputs were evaluated against ERA5-Land during the independent testing period using the common evaluation framework described in Section 2.5. Only valid paired land grid cells were included in the assessment.

The analysis includes domain-averaged performance metrics, spatial skill patterns, inter-model variability and distributional diagnostics. R, RMSE, MAE, bias, NSE and KGE were used to quantify overall agreement with ERA5-Land. Spatial maps were used to assess regional variations in correction performance, while Q–Q plots and selected-percentile analyses were used to evaluate distributional agreement and performance in the upper and lower tails.

3.1 Overall performance of bias-correction methods

Table 3 summarises the mean grid cell performance of the raw and bias-corrected CMIP6 ensemble means during the evaluation period. Each of the 20 CMIP6 models was corrected individually using QDM, RF and U-Net, after which the model outputs were averaged to obtain an ensemble mean for each method.

For precipitation, the raw CMIP6 ensemble showed an R of 0.170, RMSE of 48.04 mm.month⁻¹, MAE of 36.97 mm.month⁻¹, NSE of -2.570 and KGE of -0.058. QDM reduced RMSE and MAE to 42.40 and 32.27 mm.month⁻¹, respectively, while R increased to 0.230 and KGE to 0.210. RF produced an R of 0.380, RMSE of 31.30 mm.month⁻¹, MAE of 23.97 mm.month⁻¹, NSE of 0.079 and KGE of 0.250. The U-Net-corrected ensemble achieved the highest precipitation R, NSE and KGE values of 0.689, 0.380 and 0.550, respectively. It also reduced RMSE to 26.93 mm.month⁻¹ and MAE to 20.16 mm.month⁻¹. Relative to the raw ensemble, this represents a 43.9% reduction in RMSE and a 45.5% reduction in MAE.

For surface soil moisture, the raw ensemble produced an R of 0.464, RMSE of 10.694 mm, MAE of 9.327 mm, NSE of -58.926 and KGE of -0.324. QDM increased R to 0.507 and reduced RMSE and MAE to 4.518 and 3.526 mm, respectively. RF further increased R to 0.669, NSE to 0.397 and KGE to 0.553, while reducing RMSE to 3.338 mm and MAE to 2.523 mm. U-Net achieved the highest soil-moisture R, NSE and KGE values of 0.750, 0.451 and 0.598, respectively, together with the lowest RMSE of 3.065 mm and MAE of 2.304 mm. Compared with the raw ensemble, U-Net reduced RMSE and MAE by 71.3% and 75.3%, respectively.

Surface runoff showed the lowest raw correlation and efficiency values. The raw ensemble produced an R of 0.164, RMSE of 14.377 mm.month⁻¹, MAE of 8.831 mm.month⁻¹, NSE of -555.774 and KGE of -11.178. QDM increased R to 0.275 and reduced RMSE and MAE to 4.035 and 2.281 mm.month⁻¹, respectively. RF produced the lowest runoff RMSE and MAE values of 0.350 and 0.221 mm.month⁻¹, respectively, with a small mean bias of -0.024 mm.month⁻¹. Its R, NSE and KGE also increased to 0.505, 0.157 and 0.257, respectively, although these values remained lower than those obtained using U-Net. U-Net achieved the highest runoff R, NSE and KGE values of 0.706, 0.386 and 0.571, respectively, with an RMSE of 2.425 mm.month⁻¹ and an MAE of 1.279 mm.month⁻¹. Relative to the raw ensemble, U-Net reduced runoff RMSE by 83.1% and MAE by 85.5%.

Across the three variables, U-Net consistently produced the highest R, NSE and KGE values. It also produced the lowest precipitation and soil-moisture RMSE and MAE, whereas RF produced the lowest reported runoff RMSE and MAE.

Table 3. Spatial means of grid-cell temporal performance metrics for the raw and bias-corrected CMIP6 multi-model ensemble fields relative to ERA5-Land during the January 2008–December 2014 evaluation period. Higher R, NSE and KGE and lower RMSE, MAE and absolute bias indicate better performance.

Variable	Method	R	RMSE	MAE	Bias	NSE	KGE
Precipitation	Raw CMIP6	0.17	48.04	36.97	2.51	-2.57	-0.058
	QDM corrected	0.23	42.4	32.27	1.04	-0.553	0.21
	RF corrected	0.38	31.3	23.97	-0.28	0.079	0.25
	U-net corrected	0.689	26.93	20.16	1.22	0.38	0.55
Surface Runoff	Raw CMIP6	0.164	14.377	8.831	5.639	-555.774	-11.178
	QDM corrected	0.275	4.035	2.281	-0.052	-14.225	0.245
	RF corrected	0.505	0.350	0.221	-0.024	0.157	0.257
	U-net corrected	0.706	2.425	1.279	-0.3181	0.386	0.571
Soil Moisture	Raw CMIP6	0.464	10.694	9.327	-1.814	-58.926	-0.324
	QDM corrected	0.507	4.518	3.526	0.1802	-0.42	0.431
	RF corrected	0.669	3.338	2.523	0.515	0.397	0.553
	U-net corrected	0.75	3.065	2.304	0.519	0.451	0.598

3.2 Spatial distribution of changes in grid-cell temporal correlation

Figure 4 shows the spatial changes in temporal correlation following bias correction relative to the raw CMIP6 simulations. At each valid land grid cell, ΔR was calculated as the correlation of the corrected monthly series with ERA5-Land minus the corresponding correlation of the raw CMIP6 series. The reported mean ΔR values represent the spatial average of these grid-cell differences across the European land domain during the independent testing period. These maps describe the spatial distribution of changes in temporal correlation at individual grid cells. They do not measure spatial dependence or spatial coherence between neighbouring grid cells.

For precipitation, QDM produced relatively small correlation changes, with positive values mainly across central, eastern and south-eastern Europe and near-zero or negative changes over parts of western Europe and northern Scandinavia. The spatial mean ΔR was 0.067. RF produced positive changes over a larger proportion of the domain, particularly across central, eastern and northern Europe, with a mean ΔR of 0.176. U-Net produced widespread positive changes across most European land areas, with relatively few grid cells showing reduced correlation. Its mean ΔR was 0.483.

For surface soil moisture, the spatial changes were more heterogeneous. QDM produced positive correlation changes across northern Europe, Scandinavia and parts of the Alpine region, while near-zero or negative changes occurred over parts of western, southern and eastern Europe. The mean ΔR was 0.091. RF produced a similar spatial pattern, with a mean ΔR of 0.120. U-Net increased correlation across a larger part of northern, central and eastern Europe, although weak or negative changes remained over some areas of western and southern Europe. The mean ΔR for U-Net was 0.274.

For surface runoff, QDM produced positive changes across much of central, northern and eastern Europe, with weaker or negative changes over some western and southern regions. The mean ΔR was 0.166. RF showed a broadly similar pattern and produced a mean ΔR of 0.292. U-Net produced positive correlation changes across most of the European land domain, particularly over central and eastern Europe, with only limited areas of weak or negative change. Its mean ΔR was 0.537. Across all three variables, U-Net produced the largest spatial mean increase in grid-cell correlation, followed by RF and QDM.

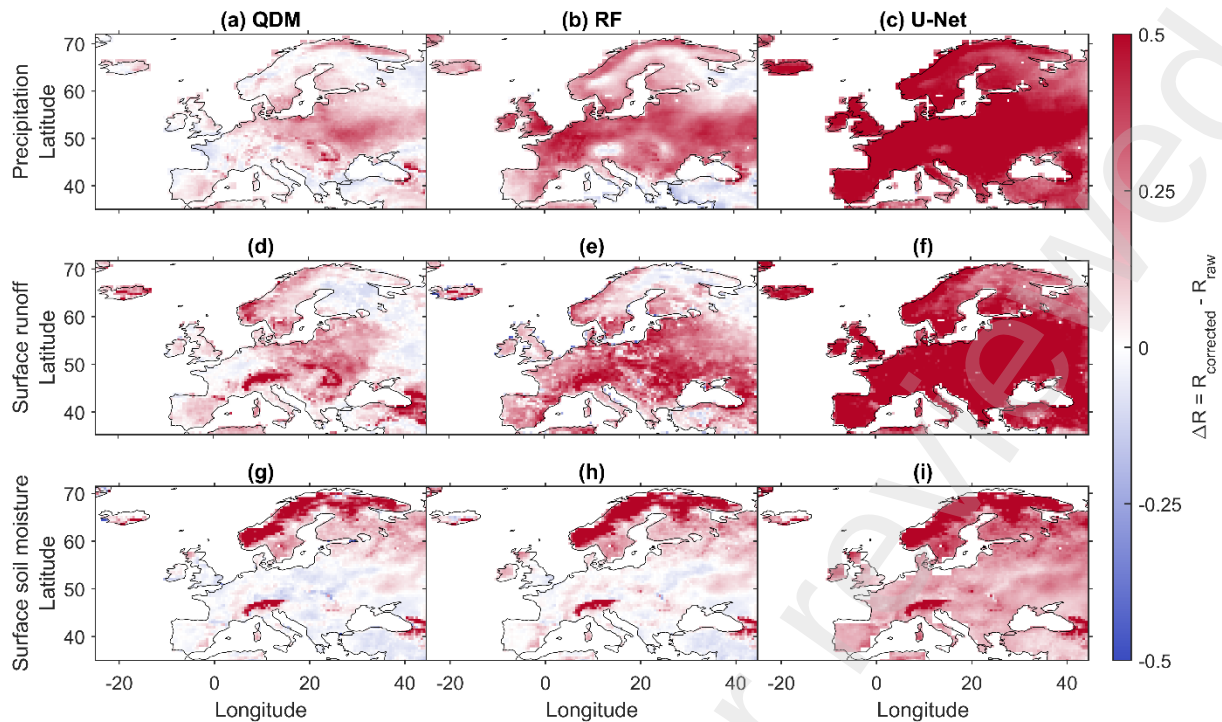


Figure 4. Spatial changes in temporal correlation following bias correction for precipitation, surface soil moisture and surface runoff during the independent testing period. At each valid European land grid cell, ΔR was calculated as $R_{\text{corrected}} - R_{\text{raw}}$ using the monthly time series. Positive values indicate increased correlation with ERA5-Land, whereas negative values indicate reduced correlation.

3.3 Inter-model uncertainty and consistency of bias-correction performance

Inter-model behaviour was examined using three complementary temporal diagnostics. Figure 5 compares the temporal correlation values obtained for the 20 individual CMIP6 models during the independent testing period. Figure 6 presents the temporal inter-model spread over 1980–2014 using the 5th–95th and 25th–75th percentile ranges around the multi-model median. Figure 7 uses Taylor diagrams to compare temporal correlation, normalized standard deviation and centred RMSE with ERA5-Land. These diagnostics evaluate temporal performance and model-to-model dispersion; they do not directly assess spatial dependence or spatial coherence.

Figure 5 shows that the bias-correction methods produced different degrees of improvement in temporal correlation. For precipitation, the raw simulations were characterized by generally low correlations. QDM produced a modest improvement, whereas RF shifted the distribution towards higher values. U-Net produced the highest correlations and the clearest separation from the other methods. A similar pattern was observed for surface soil moisture, although the raw simulations already showed moderate temporal agreement with ERA5-Land. QDM and RF increased the correlations, while U-Net produced the highest and most compact distribution. For surface runoff, the raw simulations showed low and relatively variable correlations. QDM and RF improved the correlations but retained some inter-model variability, whereas U-Net produced consistently higher values, ranging approximately from 0.62 to 0.81.

Figure 6 shows a marked reduction in temporal spread following bias correction, indicating a corresponding decrease in inter-model uncertainty. The raw simulations exhibit the broadest 5th–95th and 25th–75th percentile ranges for all three variables, with particularly large uncertainty in surface runoff and surface soil moisture. For precipitation, the raw ensemble also shows substantial year-to-year variation in both the median and percentile envelopes, indicating that the magnitude of uncertainty changes through time. QDM considerably narrows the uncertainty ranges, especially for soil moisture and runoff, although precipitation retains a visible spread and temporal variation in the outer percentile band. RF produces similarly compact distributions, with narrow central ranges and relatively stable

multi-model medians. Its precipitation spread is generally smaller than that of QDM, while the runoff envelope remains slightly variable between years, indicating that some time-dependent uncertainty persists after correction.

U-Net produces the most concentrated model distributions, particularly for surface soil moisture and runoff, where the interquartile ranges closely follow the multi-model median. The U-Net precipitation series retains distinct interannual fluctuations, including several pronounced increases and decreases, but these changes occur within a substantially narrower ensemble range than in the raw simulations. Surface runoff shows the largest absolute reduction in uncertainty, with the broad raw envelope contracting sharply under all three correction methods. Soil moisture also shows a strong reduction, changing from a wide raw range to closely clustered corrected values. The wider 5th–95th percentile bands relative to the 25th–75th percentile ranges indicate that the central group of corrected models remains closely clustered around the median, while greater differences persist toward the outer parts of the ensemble distribution.

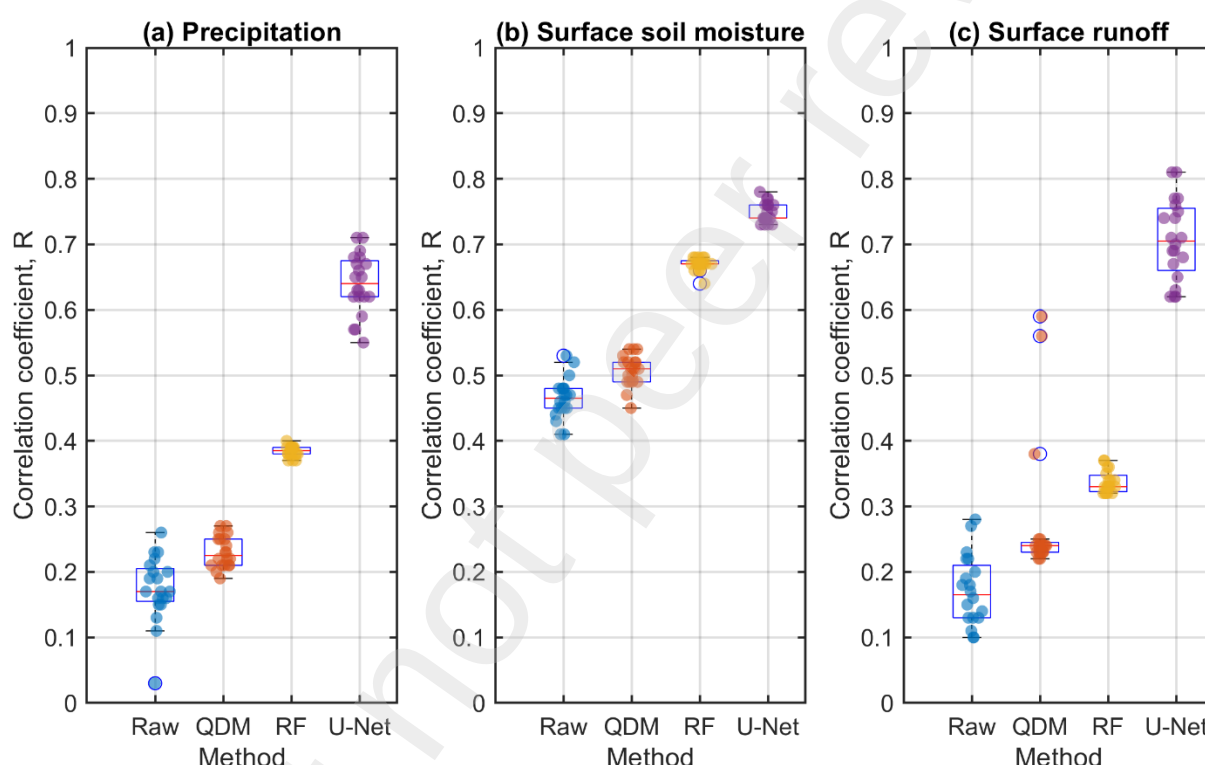


Figure 5. Distribution of temporal correlation coefficients across the 20 CMIP6 models for precipitation, surface soil moisture and surface runoff during the testing period. Each point represents an individual CMIP6 model.

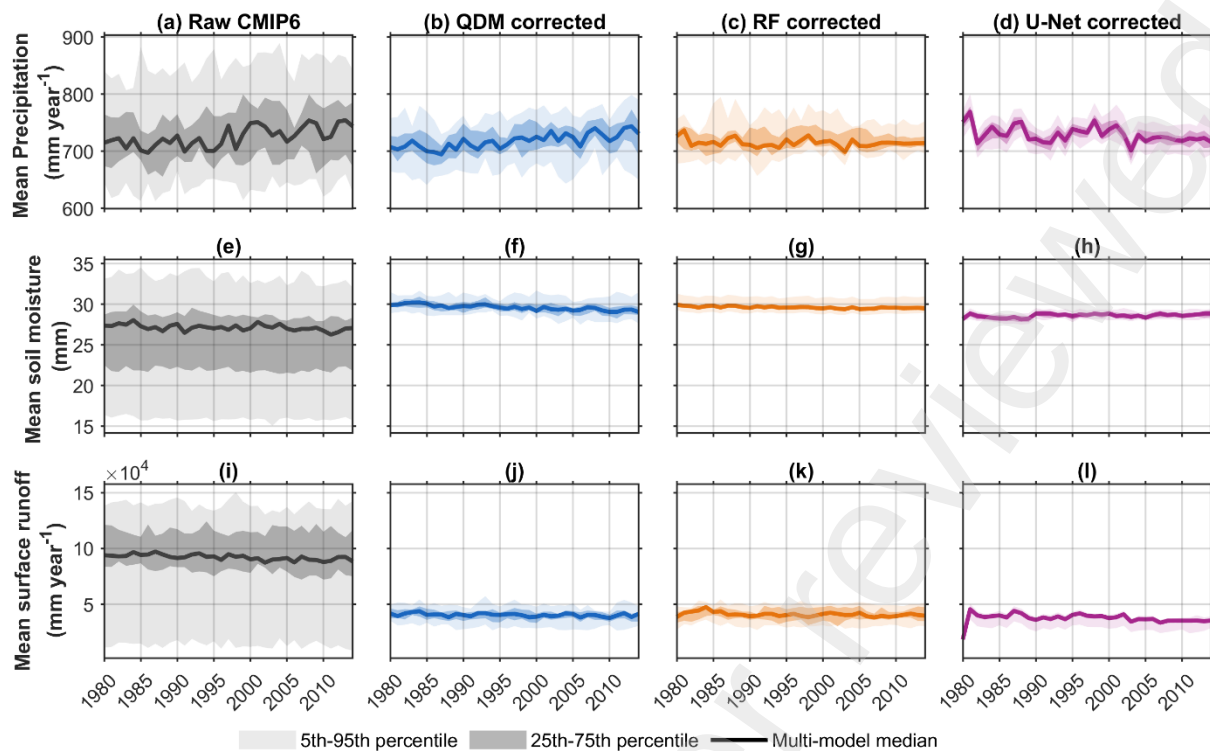


Figure 6. Temporal inter-model spread of precipitation, surface soil moisture and surface runoff during 1980–2014 for the raw, QDM-, RF- and U-Net-corrected simulations. The light and dark shaded areas represent the 5th–95th and 25th–75th percentile ranges, respectively, and the solid line represents the multi-model median.

Figure 7 compares the temporal performance of the 20 CMIP6 models using correlation, normalized standard deviation and centred RMSE relative to ERA5-Land. For precipitation, the raw simulations were concentrated at low correlation values and showed visible inter-model spread in normalized standard deviation. QDM produced a narrower cluster and reduced the variability spread, although correlation remained relatively low. RF shifted the models towards higher correlation and lower centred RMSE. U-Net produced the clearest separation from the raw and QDM simulations, with the individual models concentrated at higher correlation values and the method mean positioned closest to the ERA5-Land reference point.

For surface soil moisture, the raw models showed the widest dispersion in both correlation and normalized standard deviation. QDM reduced the spread among the individual models and produced a more compact cluster. RF increased correlation and shifted the models towards lower centred RMSE whereas U-Net produced the highest correlation values and the tightest clustering, with its method mean located closest to ERA5-Land.

Surface runoff showed the largest difference between the raw and corrected simulations. The raw models were distributed at low correlation values and showed substantial inter-model variation in normalized standard deviation and centred RMSE. QDM reduced the spread in normalized standard deviation, while RF produced a further shift towards lower centred RMSE and moderately higher correlation. U-Net formed the most compact runoff cluster and produced the highest correlations, with its method mean positioned closest to the ERA5-Land reference. Across all three variables, the raw simulations showed the greatest inter-model dispersion. QDM mainly narrowed the spread in normalized standard deviation, RF produced broader improvements in correlation and centred RMSE, and U-Net showed the closest overall clustering towards ERA5-Land.

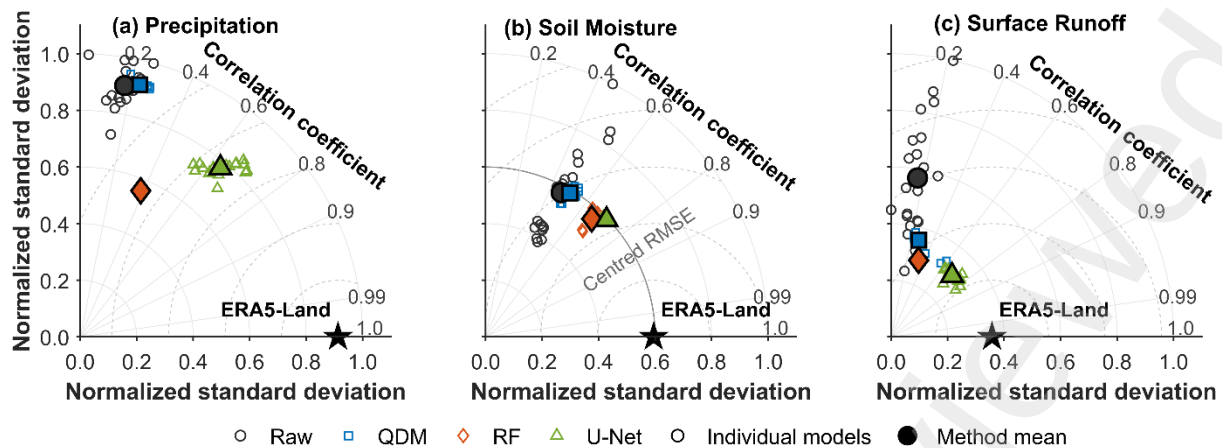


Figure 7. Taylor diagrams comparing the temporal performance of the raw, QDM-, RF- and U-Net-corrected CMIP6 simulations with ERA5-Land during the independent testing period. The diagrams summarize temporal correlation, normalized standard deviation and centred RMSE for precipitation, surface soil moisture and surface runoff.

3.4 Distributional performance of bias-correction methods

Figure 8 compares the distributions of raw and bias-corrected precipitation, surface soil moisture and surface runoff with ERA5-Land using Q–Q plots, mean absolute quantile error (MAQE) and selected upper-tail quantile biases. The proximity of each curve to the 1:1 line shows the level of agreement between the simulated and ERA5-Land quantiles across the full distribution.

For precipitation, the raw CMIP6 ensemble remained relatively close to the 1:1 line across the lower and middle quantiles but increasingly deviated from ERA5-Land towards the upper tail. The negative Q95 and Q99 biases show that the highest precipitation quantiles were underestimated. QDM produced the closest overall alignment with the reference distribution and reduced MAQE to $0.15 \text{ mm.month}^{-1}$. Its curve remained close to the 1:1 line across most of the quantile range, with only small positive deviations at the highest quantiles. RF showed a greater departure from ERA5-Land in the upper part of the distribution and retained marked negative Q95 and Q99 biases. U-Net closely reproduced the lower and central quantiles and remained near the reference line through most of the distribution, although a small positive deviation appeared at the highest precipitation values.

For surface soil moisture, the raw CMIP6 distribution showed a narrower quantile range than ERA5-Land. The lower quantiles were generally overestimated, whereas the divergence increased progressively towards the wetter part of the distribution. The raw Q99 bias reached -27.10 mm , representing the largest upper-tail soil-moisture error among the evaluated datasets. QDM produced close agreement with ERA5-Land across nearly the full distribution, reduced MAQE to 0.49 mm and left only a small upper-tail bias. RF reduced the overall departure from the raw distribution but continued to underestimate the higher soil-moisture quantiles. U-Net remained close to the 1:1 line across the lower, central and moderately high quantiles, with a small overestimation at the wettest end of the distribution.

Surface runoff showed the largest raw distributional departure from ERA5-Land. The raw ensemble had an MAQE of $5.69 \text{ mm.month}^{-1}$ and overestimated values across much of the lower and central quantile range. QDM substantially reduced these differences, lowering MAQE to $0.48 \text{ mm.month}^{-1}$ and closely following the reference line over most quantiles. RF also reduced the raw distributional error but showed increasing negative deviation in the upper tail, including a negative Q99 bias. U-Net produced the lowest runoff MAQE of $0.23 \text{ mm.month}^{-1}$ and closely reproduced the lower and central quantiles. A moderate negative deviation remained at the highest runoff values.

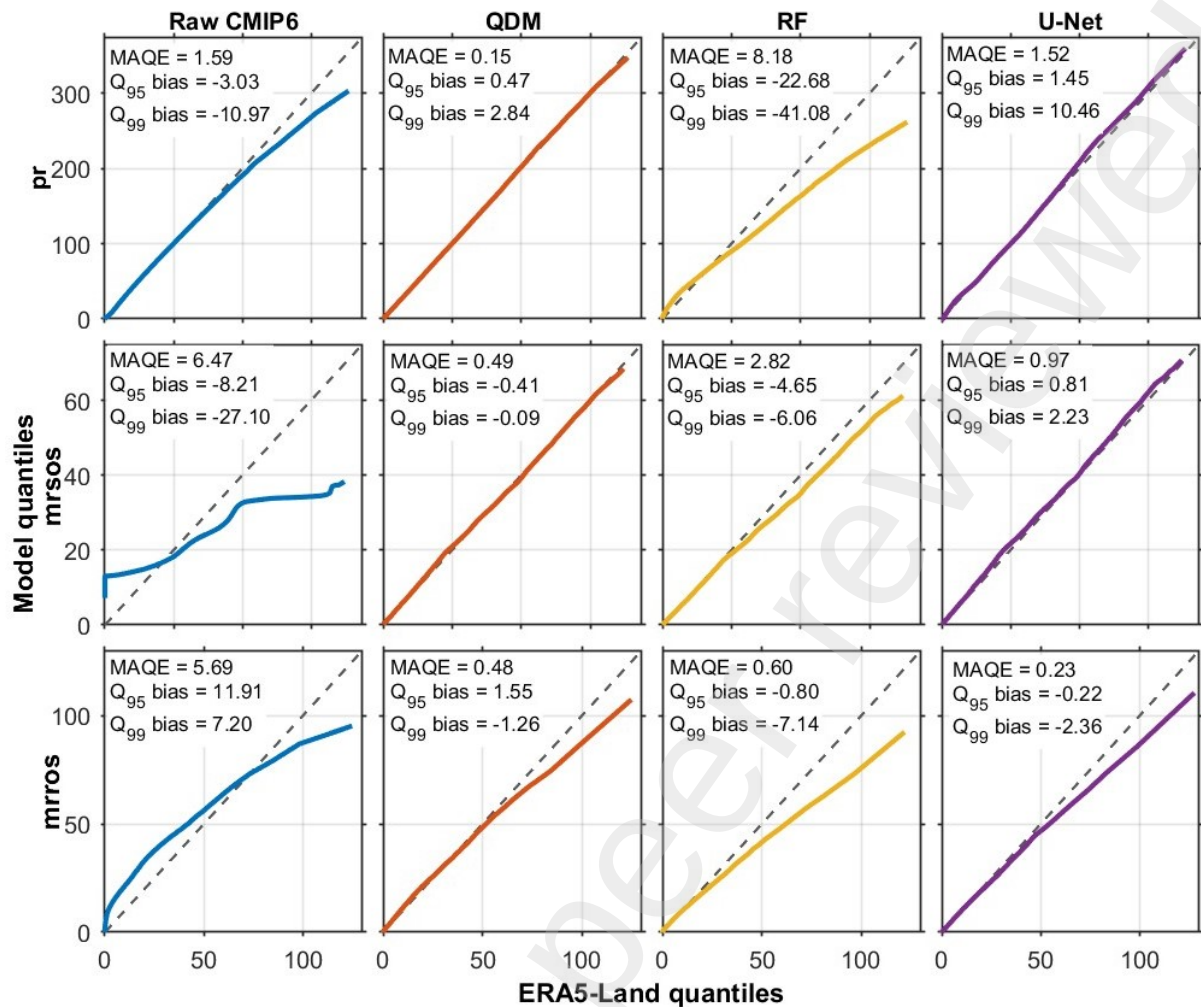


Figure 8. Q-Q comparison of raw, QDM-, RF- and U-Net-corrected precipitation, surface soil moisture and surface runoff against ERA5-Land during the independent testing period. The dashed 1:1 line represents perfect quantile agreement. MAQE denotes mean absolute quantile error, while Q95 and Q99 biases represent model minus ERA5-Land quantiles.

3.5 Performance in extreme percentiles

Figure 9 presents the spatial bias in selected extreme percentiles for raw and bias-corrected precipitation, surface soil moisture and surface runoff relative to ERA5-Land. Precipitation and surface runoff were evaluated using the 95th percentile, whereas surface soil moisture was assessed using the 5th percentile to represent dry-tail conditions. Positive values indicate overestimation of the selected percentile, while negative values indicate underestimation.

For precipitation P95, the raw CMIP6 ensemble showed a heterogeneous spatial pattern, with positive and negative biases distributed across Europe and a domain-mean bias of 2.25 mm.month⁻¹. Positive biases were more evident over parts of western and southern Europe, while negative biases occurred across sections of northern, central and eastern Europe. QDM reduced the magnitude of both positive and negative biases across most of the domain and produced a near-neutral mean bias of 0.23 mm.month⁻¹. RF showed widespread underestimation, particularly across northern, western and central Europe, resulting in the largest negative mean bias of -30.57 mm.month⁻¹. U-Net reduced the RF underestimation but retained negative biases over parts of northern and eastern Europe, with a mean bias of -6.46 mm.month⁻¹. Local positive biases remained over parts of western and southern Europe.

For surface soil moisture P5, the raw ensemble showed a mixed spatial pattern, with widespread negative biases and a domain-mean value of -3.57 mm. After correction, the bias pattern shifts toward positive values over much of Europe. QDM, RF and U-Net produce mean P5 biases of 2.59, 3.10 and

1.99 mm, respectively, showing that the corrected simulations generally estimate slightly higher P5 soil moisture values than ERA5-Land. Among the corrected methods, U-Net gives the smallest positive mean bias, while RF shows the strongest positive P5 bias. Both methods reduced the magnitude of the raw negative bias, although regional overestimation remained.

For surface runoff P95, the raw ensemble showed strong positive bias over northern and north-eastern Europe and negative bias over parts of southern Europe and the Alpine region. The domain-mean bias was 12.96 mm.month⁻¹. QDM, RF and U-Net substantially reduced the raw bias, producing mean values of -2.94, -2.15 and -2.45 mm.month⁻¹, respectively. The corrected maps showed smaller spatial bias magnitudes than the raw ensemble, although negative residual biases remained over parts of central, southern and south-eastern Europe. Among the corrected methods, RF produced the smallest absolute domain-mean runoff P95 bias, followed by U-Net and QDM. The selected-percentile maps, therefore, show that bias correction substantially reduces the large spatial errors in the raw CMIP6 simulations, but the remaining biases differ by variable, method and region.

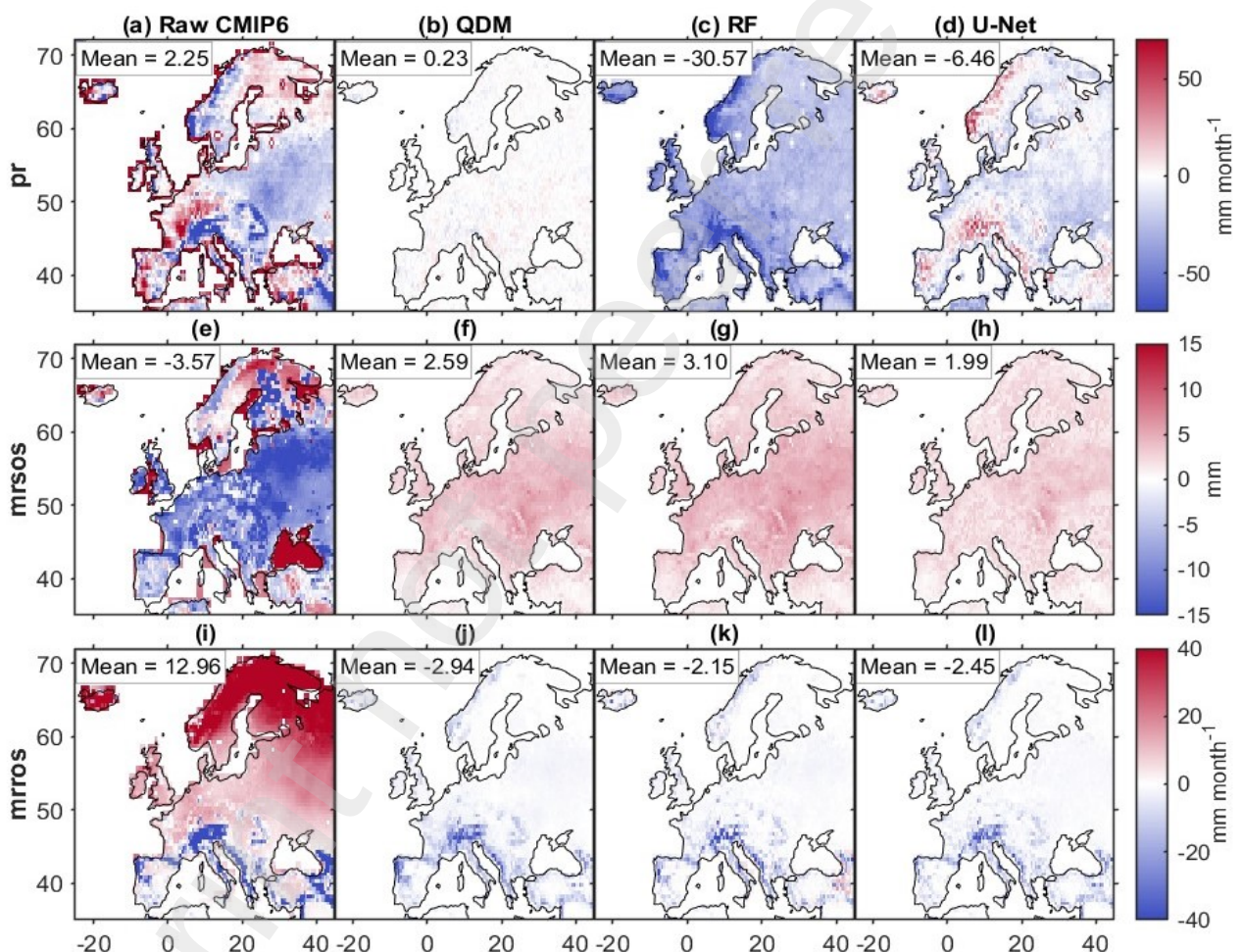


Figure 9. Spatial bias in selected extreme percentiles relative to ERA5-Land during the independent testing period. Maps show model percentile minus ERA5-Land percentile for raw, QDM-, RF- and U-Net-corrected simulations. Precipitation and surface runoff are evaluated using P95, while surface soil moisture is evaluated using P5. Positive values indicate overestimation and negative values indicate underestimation.

3.6 Model ranking across bias-correction methods

Figure 10 presents the TOPSIS rankings of the 20 CMIP6 models based on their combined KGE and RMSE performance. Rankings were calculated separately for precipitation, surface soil moisture and surface runoff and independently for the raw, QDM-, RF- and U-Net-corrected simulations. Clear differences in model order occurred among the correction methods, showing that the relative performance of individual models changed after bias correction.

For precipitation, the raw and QDM rankings retained several common models within the leading groups, indicating comparatively limited reorganisation after distribution-based correction. However, changes were still visible within the first five and first ten positions. RF produced a stronger redistribution of the ranking order, with several models moving between the upper, middle and lower groups. The largest changes occurred after U-Net correction. Models that had moderate or relatively weak raw rankings moved into the leading U-Net positions, while some models that performed strongly in the raw simulation shifted downward after correction. CMCC-CM2-HR4 ranked first under U-Net, whereas CESM2 remained among the higher-performing precipitation models across more than one dataset. The precipitation ranking therefore changed not only at the first position but across a wider part of the model ensemble.

Surface soil moisture showed the greatest overall reorganisation. The highest-ranked raw model did not retain first position after correction, and the composition of the leading five models changed markedly among QDM, RF and U-Net. Under U-Net, CMCC-ESM2 moved to first position despite ranking lower in the raw and other corrected datasets. Several models outside the leading raw group also entered the top five or top ten following correction. Conversely, some models with favourable raw rankings moved to lower positions. These shifts produced a substantially different post-correction model hierarchy and show that the combined KGE–RMSE performance of soil-moisture simulations was strongly affected by the selected correction method.

Surface runoff showed greater ranking persistence. MRI-ESM2-0 remained within the first two positions across all datasets and ranked first for the raw, QDM and RF simulations. Several other runoff models also remained within similar leading groups across multiple methods. U-Net produced some reordering, with UKESM1-0-LL moving to first position, but the changes were less extensive than those observed for precipitation and soil moisture. The runoff results therefore contained both stable high-performing models and method-dependent changes near the top of the ranking. The result therefore show that model selection based only on raw simulations may not represent post-correction performance, particularly for precipitation and surface soil moisture. Model ranking should therefore be conducted separately for each hydro-climatic variable and after applying the intended bias-correction method.

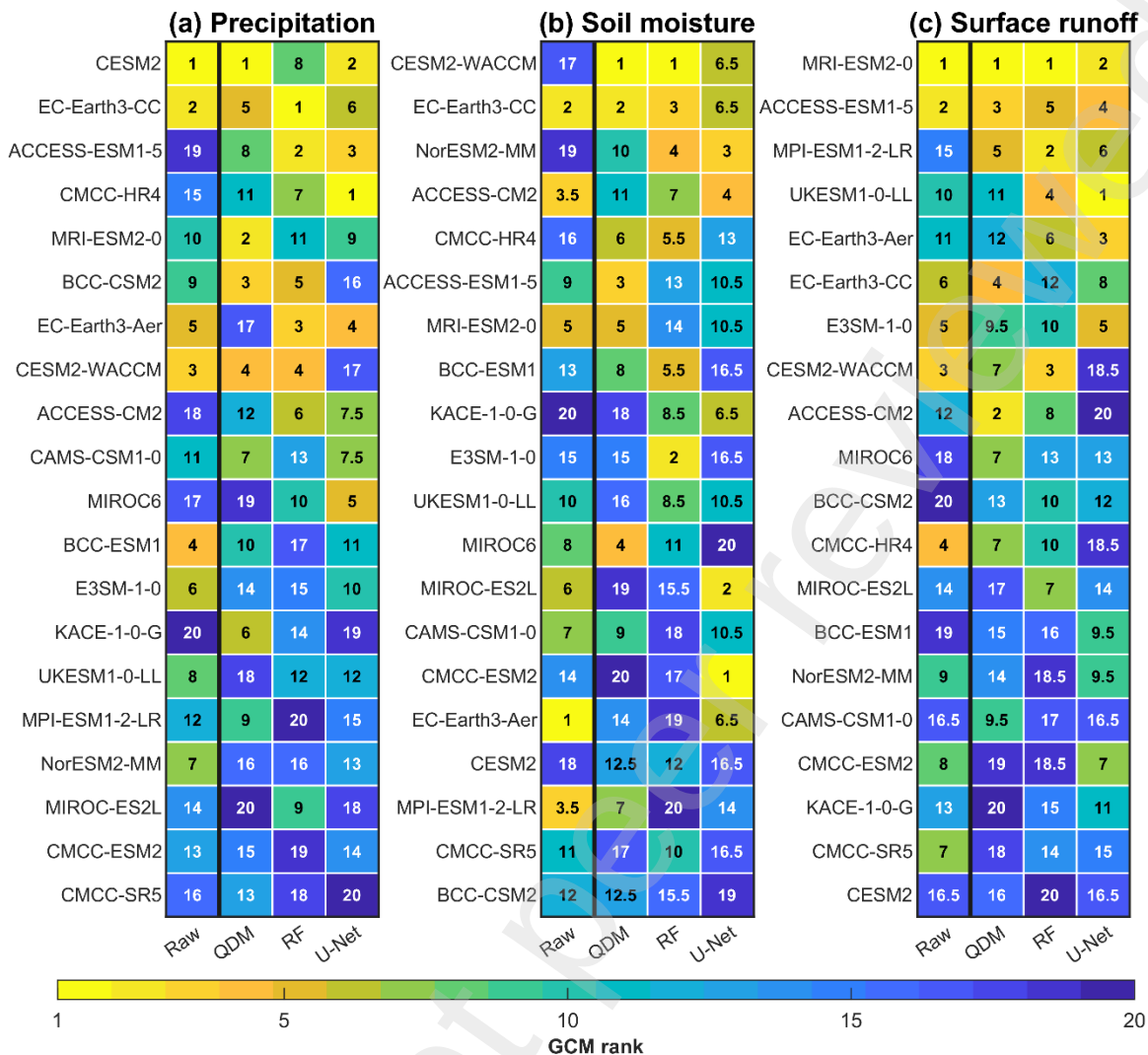


Figure 10. TOPSIS-based ranking of the 20 CMIP6 models for precipitation, surface soil moisture and surface runoff under raw, QDM-, RF- and U-Net-corrected conditions. Rankings combine KGE as a benefit criterion and RMSE as a cost criterion using weights of 0.60 and 0.40, respectively. Lower rank numbers indicate stronger combined performance.

4. Discussion

Climate-model outputs contain systematic errors that can limit their direct use in regional hydro-climatic assessments. The effectiveness of a correction method depends on the target variable, reference dataset, spatial and temporal scale, evaluation metric and intended application^{4,6,54}. Distributional agreement, spatial pattern agreement and time-series statistics describe different aspects of performance and do not necessarily improve together^{4,11,54}. Recent evaluation frameworks therefore emphasise that bias-adjustment methods should be assessed against the properties that matter for the intended application, including marginal distributions, spatial dependence, temporal structure and changes in the simulated climate signal⁵⁵.

In this study, QDM, RF and variable-specific U-Net approaches generally improved agreement between CMIP6 and ERA5-Land, although their performance varied across precipitation, surface soil moisture and surface runoff. QDM showed the closest agreement for several distributional and quantile-based diagnostics. RF generally reduced magnitude-based errors and improved correlation compared with QDM. The selected U-Net framework achieved the highest R, NSE and KGE values and produced the most widespread increases in grid-cell correlation. Overall, the results show that no single metric is sufficient to evaluate bias-correction performance and that the relative strengths of each method depend on the variable and diagnostic considered. However, these improvements represent closer agreement

with ERA5-Land and should not be interpreted as independent evidence of improved observational or hydrological accuracy.

4.1 Comparative performance of bias-correction methods

The methods showed different strengths in distributional adjustment and overall agreement with ERA5-Land. QDM closely reproduced the precipitation and surface soil-moisture quantiles and substantially reduced the distributional error of surface runoff, as indicated by the Q-Q plots in Figure 8. This behaviour is consistent with the design of QDM, which adjusts quantile-dependent distributional differences while seeking to preserve modelled changes in the corresponding quantiles¹¹.

The increases in monthly correlation under QDM were comparatively small. For precipitation, RMSE decreased from 48.04 to 42.40 mm.month⁻¹, while R increased from 0.170 to 0.230 (Table 3). The spatial mean changes in grid-cell correlation were also smaller for QDM than for RF or the U-Net method (Figure 4). However, this should not be interpreted as evidence that QDM failed to recover the timing of observed monthly anomalies. The correlations were calculated from monthly values that retained the seasonal cycle, and seasonalised anomaly correlations, event-coincidence, persistence or lag diagnostics were not reported.

More fundamentally, standard CMIP6 historical simulations are free-running climate data. External forcings constrain their statistical climate response, but internal variability does not generally follow the observed sequence of individual monthly weather and climate anomalies. Same monthly correlation with ERA5-Land can therefore reflect common seasonality, spatial climatology, forced low-frequency changes and other statistical structures, but it should not be described as recovered event timing or temporal sequencing. This limitation has been emphasised in studies showing that direct time-step comparison of free-running climate simulations with observed sequences can be dominated by internal variability^{4,56,57}.

RF generally produced stronger monthly correlations and lower average errors than QDM, particularly for precipitation and surface soil moisture (Table 3). The RF model used the simulated value, latitude and longitude as predictors, allowing it to learn nonlinear relationships between CMIP6 and ERA5-Land that varied across space. Similar benefits of RF-based bias correction have been reported for CMIP6 precipitation, particularly for capturing nonlinear model-reference relationships and reducing systematic errors^{15,20}. However, the improvements were less consistent across R, NSE and KGE, and the increases in grid-cell temporal correlation shown in Figure 4 were more spatially heterogeneous than those obtained with the U-Net-based configurations. RF was applied to individual grid-cell samples and did not directly use information from surrounding grid cells. This may partly explain differences in the spatial distribution of grid-cell performance, although the present analysis cannot separate the effect of neighbourhood information from other differences in preprocessing, predictors, temporal information and model configuration.

The increases in grid-cell temporal correlation under RF were less extensive and more spatially heterogeneous than those obtained under the U-Net-based method (Figure 4). For surface runoff, RF produced an R of 0.505, RMSE of 0.350 mm month⁻¹, MAE of 0.221 mm month⁻¹, bias of -0.024 mm month⁻¹, NSE of 0.157 and KGE of 0.257. These results show that RF substantially reduced the magnitude of runoff errors and improved temporal agreement and hydrological efficiency relative to the raw and QDM-corrected simulations. However, U-Net retained higher R, NSE and KGE values, indicating stronger performance across the correlation and efficiency measures. The two methods therefore showed different strengths: RF provided the lowest average-error metrics, whereas U-Net achieved the more favourable overall balance of temporal agreement, variability and bias. Similar disagreement between magnitude-based and time-series-based metrics has been reported in hydrological bias-correction studies, where methods that reproduced mean values or statistical distributions did not necessarily achieve the strongest temporal correlation or NSE^{6,58}.

Among the evaluated configurations, the variable-specific U-Net method produced the highest reported domain-level R, NSE and KGE values relative to ERA5-Land. Table 3 shows the highest domain-level values for all three variables, while Figure 4 shows the largest spatial mean increases in grid-cell correlation: 0.483 for precipitation, 0.274 for surface soil moisture and 0.537 for surface runoff. The Taylor diagrams in Figure 7 further show that the corrected model sets generally shifted towards higher correlation, lower centred RMSE and improved normalised variability relative to ERA5-Land. Similar improvements have been reported in deep-learning climate-correction studies in which convolutional architectures used spatially organised information to improve precipitation fields^{16,59}. CNN-based model-error correction has also improved aspects of mean bias and variability in other experimental settings⁶⁰. These studies demonstrate the broader potential of spatial deep-learning methods, although differences in datasets, temporal resolution, correction objectives and model design limit direct comparison with the present variable-specific configurations.

The U-Net-based framework was not superior for every distributional diagnostic. QDM produced the lowest precipitation and soil-moisture MAQE values in Figure 8, while departures remained in the highest monthly quantiles after U-Net-based correction. Figure 9 also shows differences among the methods in the spatial biases of precipitation P95, runoff P95 and soil-moisture P5. The largest overall improvement was observed for surface runoff, which had the weakest uncorrected agreement with ERA5-Land. The selected runoff configuration combined QDM preprocessing, seven-month lagged anomalies, seasonal training and residual field prediction. These components may have helped reproduce the statistical structure of ERA5-Land surface-runoff fields, although the present analysis cannot determine the contribution of each component separately.

However, the present results show that the complete U-Net-based configurations were effective, rather than isolating the contribution of the U-Net architecture alone. The precipitation configuration combined QDM preprocessing, seasonal stratification, direct field prediction and a compound loss. Soil moisture used a separate seasonal lagged anomaly formulation with three-month temporal input and MAE loss, while runoff used seven-month lagged anomalies, seasonal stratification and residual learning relative to a QDM baseline. The observed advantage should therefore not be attributed solely to greater model complexity or spatial learning, because the configurations also differed in preprocessing, temporal inputs, prediction formulation and loss function. A controlled ablation analysis would be required to determine the individual contribution of each component.

4.2 Hydro-climatic controls on bias-correction performance

Differences in bias-correction performance reflect the contrasting statistical and physical behaviour of variables as well as the relationship to ERA5-Land reference. However, the results should not be used to infer physical mechanisms that were not directly represented or evaluated. Their unequal temporal persistence and spatial organisation mean that the same method can correct one variable effectively while providing weaker improvement for another.

For precipitation, climate-model errors are multidimensional, involving biases not only in rainfall frequency and magnitude but also in the timing, persistence and spatial organisation of precipitation systems. QDM closely reproduced most of the reference quantile distribution (Figure 8), consistent with its direct treatment of monthly, grid-cell-specific empirical distributions. The limited increase in monthly correlation does not show that precipitation was assigned to the wrong observed month because the free-running CMIP6 simulation were not expected to reproduce the observed sequence of internal anomalies. Strong quantile agreement can therefore coexist with limited improvement in monthly correlation. This distinction is consistent with previous bias-correction studies showing that improvement in marginal distributions does not necessarily produce equivalent improvement in temporal association or spatial dependence^{4,11}. However, the present analysis did not directly evaluate de-seasonalised anomaly correlation or spatial coherence.

The U-Net-based precipitation configuration produced widespread increases in grid-cell temporal correlation in Figure 4, higher correlations across the individual CMIP6 models in Figure 5 and more favourable Taylor-diagram statistics in Figure 7. Its formulation allowed surrounding grid values to be used during prediction, unlike QDM and the point-sample RF formulation. This may partly explain the more widespread increases in grid-cell temporal correlation. However, the configuration also included QDM preprocessing, elevation, separate seasonal models and a combined MAE, correlation and variance loss. The present experiment therefore cannot isolate whether the improvements resulted from convolutional neighbourhood information, preprocessing, auxiliary predictors, seasonal training or their combined effects. Previous studies show that spatial deep-learning methods can improve precipitation correction by using gridded information^{16,59,61,62}, but differences in model design, data resolution and evaluation framework limit direct comparison with the present configuration.

The remaining departures at Q95 and Q99 in Figure 8 and the regional P95 biases in Figure 9 indicate residual errors in the upper tail of monthly precipitation totals. Because the testing period contains only 84 months, the grid-cell Q99 estimates are determined by very few observations and are subject to substantial sampling uncertainty. These diagnostics therefore describe monthly upper-tail behaviour rather than daily precipitation extremes, storm intensity or return levels.

Previous studies have shown that deep-learning models may retain errors in sparsely sampled distribution tails unless the training objective gives sufficient weight to high values^{61,63}. The present results therefore show a trade-off among the evaluated configurations: QDM provided stronger marginal-distribution and selected monthly-quantile agreement, whereas the U-Net-based configuration produced larger improvements in grid-cell correlation, efficiency metrics and inter-model consistency. Method selection should consequently reflect whether the intended application prioritises marginal-distribution agreement, calendar-matched field-level association or performance in monthly upper-tail conditions. Method selection should therefore reflect whether the application prioritises distributional fidelity, regional precipitation structure or extreme-value accuracy.

Surface soil moisture showed a more persistent but spatially heterogeneous correction response because its evolution reflects both current atmospheric forcing and antecedent land-surface conditions. The strong QDM performance in Figure 8 indicates that a substantial component of the model-reference difference was associated with the marginal distribution. U-Net and RF produced higher nonlinear and spatially varying relationships according to the selected correlation and efficiency metrics, although the causes of the regional performance differences were not directly tested. Soil texture, vegetation, topography, evaporative demand, snow conditions and land-cover differences are plausible controls, but alternative explanations include regional variation in ERA5-Land uncertainty, native CMIP6 resolution and effective sample variability. Although the ERA5-Land and CMIP6 soil-moisture fields were harmonised to a common 0–10 cm depth, residual differences may remain because the CMIP6 models use different native soil-layer structures and land-surface parameterisations. The reported differences may therefore reflect both statistical bias and model-specific representation of near-surface soil moisture.

Previous studies have likewise shown that soil-moisture modelling and correction skill is sensitive to climatic setting, land cover and terrain heterogeneity, with weaker transferability across contrasting surface conditions^{12,64,65}. The weaker gains over parts of southern and western Europe may therefore reflect stronger seasonal drying, higher evaporative demand and shorter soil-moisture memory, which increase the nonlinearity and regional variability of the correction relationship. Despite this spatial heterogeneity, the narrower uncertainty envelopes in Figure 6 and tighter clustering in Figure 7 indicate that U-Net reduced both individual-model error and inter-model disagreement. Despite this spatial heterogeneity, the U-Net-based soil-moisture configuration produced the highest reported R, NSE and KGE values and reduced the inter-model spread relative to the raw simulations. These results demonstrate improved agreement with ERA5-Land according to the selected metrics, but they do not

directly establish improved spatial coherence or physical realism. The strong QDM performance in Figure 8 further shows that a substantial component of the soil-moisture difference was distributional. Overall, QDM was particularly effective in reproducing the wetness range and marginal distribution, whereas the U-Net-based configuration produced the most favourable combination of the reported correlation, efficiency and model-consistency measures.

Surface runoff showed the largest reported increases in R, NSE and KGE under the selected U-Net-based configuration. This configuration differed substantially from the other methods because it combined QDM preprocessing, seven-month lagged anomalies, seasonal training and residual field prediction. These components might help to improved agreement with ERA5-Land by providing temporal context and spatially distributed inputs, but the present analysis cannot determine the individual contribution of each component. Previous studies show that runoff-related variables are sensitive to antecedent conditions and delayed land-surface responses^{13,14,56}. However, the current results should be interpreted as correction of ERA5-Land surface-runoff fields rather than direct recovery of catchment discharge processes. RF produced very low reported RMSE and MAE values but weaker R, NSE and KGE. QDM substantially improved the runoff distribution in Figure 8 but produced smaller increases in monthly correlation than the U-Net-based configuration. Quantile-based rainfall correction can reproduce monthly totals while retaining biases in autocorrelation, wet-dry sequencing and multi-day rainfall occurrence, which subsequently affects mean and high runoff^{56,66,67}.

The widespread increases in grid-cell correlation in Figure 4, upward shift across models in Figure 5 and favourable Taylor-diagram statistics in Figure 7 show that the lagged residual U-Net configuration produced stronger agreement with ERA5-Land surface-runoff fields according to several reported metrics. Because the model used gridded inputs, it could use neighbouring field values during prediction. However, these results do not establish that it learned river-basin structure, drainage connectivity or catchment storage processes. Neither channel routing nor upstream contributing area was represented, and no basin-scale or spatial-connectivity diagnostic was performed. The improvement should therefore be described as enhanced field-level agreement with ERA5-Land surface runoff rather than improved river-flow or catchment behaviour. Residual errors remained in the upper tail of monthly runoff totals, as shown by Figure 8 and the regional P95 biases in Figure 9. Because the testing period contains only 84 monthly observations, the grid-cell P95 and especially Q99 estimates are based on few values and have considerable sampling uncertainty. They should therefore be interpreted as indicators of monthly upper-tail runoff behaviour rather than event-scale runoff extremes, flood peaks or flood-relevant thresholds.

4.3 Inter-model consistency, dispersion and model ranking

The correlation distributions in Figure 5 show that bias correction altered both the central performance and the model-to-model variability of the CMIP6 ensemble. The broad and comparatively low raw distributions for precipitation and surface runoff indicate substantial differences among models in their calendar-matched agreement with ERA5-Land. QDM produced only a modest upward shift and retained substantial overlap with the raw simulations, indicating that improved marginal-distribution agreement did not produce a comparable increase in calendar-matched monthly correlation. RF increased the median correlations, particularly for precipitation and soil moisture, but the broader runoff distribution suggests that the effectiveness of local nonlinear correction remained sensitive to the error structure of each model. U-Net produced the clearest separation from the raw and QDM outputs and raised both the median and lower quartile across all variables. Improvement in the lower-performing part of the ensemble is important because it shows that the increase in skill was not driven by only a few favourable models. The U-Net also differed from QDM and RF in preprocessing, predictors, temporal context, target formulation and loss functions, so the contribution of neighbouring information cannot be isolated from the present comparison. Previous multi-model evaluations have similarly shown that bias-correction performance varies among climate models because their initial bias structures and simulated distributions differ^{68,69}.

The percentile envelopes in Figure 6 show that the corrected simulations had narrower historical inter-model dispersion in the magnitude and temporal evolution of the variables. The broad raw envelopes reflect differences among the CMIP6 simulations, while their contraction after correction shows that the model outputs became more similar to one another and to the common ERA5-Land reference. This convergence may reflect the reduction of model-specific biases. QDM strongly narrowed the runoff and soil-moisture ranges because each model distribution was adjusted toward the same reference, while RF produced similarly compact central ranges but retained some time-dependent runoff spread. The U-Net-based outputs showed the narrowest inter-model envelopes, particularly for surface soil moisture and surface runoff, with the interquartile envelopes closely following the multi-model median. However, the remaining separation between the 5th–95th and 25th–75th percentile range indicates that some models continued to depart from the ensemble centre. This historical convergence should therefore be interpreted as improved agreement with the reference rather than complete elimination of uncertainty. Conventional bias correction can artificially contract ensemble spread when all members are adjusted toward one observational dataset, potentially suppressing part of the model information represented by the original ensemble⁷⁰. It also does not remove uncertainty associated with internal variability, future forcing or structural differences in projected climate responses.

The Taylor diagrams in Figure 7 show that the narrower inter-model dispersion was accompanied by changes in correlation, normalised standard deviation and centred RMSE, rather than convergence towards a common mean alone. For precipitation, QDM mainly reduced differences in normalised variability, while the increase in correlation remained comparatively small. RF shifted the models towards higher correlation and lower centred RMSE relative to the raw simulations. Soil moisture followed a similar progression, although its higher raw correlations reflect stronger persistence and storage memory. For runoff, the movement from widely dispersed, low-correlation raw simulations toward a tighter U-Net cluster indicates simultaneous improvement in correlation, normalized variability and centred RMSE. The corrected models nevertheless remained separated in Taylor space, confirming that bias correction reduced but did not remove their individual structural characteristics. This distinction is important because improved historical agreement does not guarantee identical hydrological responses under future conditions, and bias correction itself can introduce additional uncertainty into runoff-impact modelling^{14,56,71}.

The TOPSIS rankings in Figure 10 show that bias correction reorganised the relative ordering of the CMIP6 models under the selected decision framework. The ranking combined KGE as a benefit criterion and normalised RMSE as a cost criterion, using weights of 0.60 and 0.40, respectively. It therefore summarised performance under one specified combination of correlation, variability, bias and error magnitude rather than providing an absolute measure of model quality. The contrast among the individual evaluation metrics demonstrates why model ranking can change according to the selected criteria. Multi-criteria approaches are increasingly used for CMIP6 selection because models often perform differently across metrics, variables, seasons and precipitation characteristics^{22,52,72}. The stronger reordering under U-Net for precipitation and soil moisture indicates that correction did not benefit every model equally. Models moving from moderate or weak raw positions into the leading groups likely contained systematic discrepancies that were more responsive to the applied correction, whereas some initially strong models lost their relative advantage after common biases were reduced. Runoff rankings were more stable, with MRI-ESM2-0 remaining among the leading raw, QDM and RF models, although UKESM1-0-LL ranked first after U-Net correction. The observed reorganisation shows that model selection based on corrected outputs can differ substantially from selection based on raw simulations. However, the exact ordering is conditional on the historical ERA5-Land reference, the selected KGE and RMSE criteria, their normalisation and the assigned 0.60/0.40 weights. The rankings should therefore be interpreted as decision support under the adopted configuration rather than as a definitive ordering of model quality.

4.4 Methodological implications and study limitations

The comparison of QDM, RF and the variable-specific U-Net-based configurations shows that bias-correction methods improve different characteristics of climate-model output and should not be evaluated using a single metric. QDM was particularly effective in reproducing marginal distributions and selected monthly quantiles, while RF reduced several magnitude-based errors. The U-Net-based configurations produced the highest reported R, NSE and KGE values and the largest increases in grid-cell correlation, although these results do not directly establish improved spatial coherence or recovery of observed anomaly timing. The disagreement among RMSE, correlation, NSE, KGE and monthly-tail diagnostics confirms that low average error does not necessarily imply stronger agreement across all statistical properties. Multi-metric evaluation is therefore required, particularly when corrected precipitation, soil moisture and runoff are intended for different hydro-climatic applications^{4,6,73}. The variable-dependent results also caution against applying one correction method uniformly across the hydrological cycle because precipitation, soil moisture and runoff exhibit different distributional, temporal and spatial characteristics.

The stronger reported performance of the U-Net-based configurations indicates the potential value of field-to-field correction using gridded inputs. However, their advantage over QDM and RF should not be attributed solely to convolution or neighbouring information. The precipitation configuration also included QDM preprocessing, elevation, seasonal stratification and a compound loss, while the runoff configuration combined QDM with seven-month lagged anomalies, seasonal training and residual learning. The soil-moisture formulation differed again in its input, target and loss design. The present comparison therefore evaluates complete variable-specific framework rather than isolating the effect of U-Net architecture. Previous work has similarly shown that univariate adjustment may improve marginal distributions while leaving spatial, temporal or cross-variable dependence insufficiently corrected⁷⁴⁻⁷⁶. Controlled ablation experiments would be required to determine the contributions of spatial context, preprocessing, auxiliary predictors, temporal inputs and loss functions. Method selection should consequently be guided by the intended application. Distributional methods may remain preferable when marginal or quantile agreement is the main objective, while hybrid field-based approaches may be more suitable when several complementary metrics are considered together.

The analysis was restricted to the historical period and therefore did not evaluate whether the correction relationships remain stationary under future climate change. QDM assumes that the model-reference bias relationship estimated from the historical calibration data remains sufficiently stable, while RF and U-Net similarly depend on relationships learned from historical conditions. The chronological testing period provides an independent historical evaluation but does not demonstrate transferability to future climates, particularly where conditions extend beyond the calibration range. The reported results should therefore be interpreted as historical performance and not as evidence that the correction relationships will remain valid under climate change.

Several uncertainties remain despite the overall improvements achieved by the correction methods. A key limitation is that QDM, RF and U-Net are statistical correction approaches trained or calibrated against ERA5-Land. They can improve agreement between CMIP6 and the selected reference without explicitly representing atmospheric and hydrological processes responsible for the original biases. ERA5-Land is itself a reanalysis-based land-surface product rather than an error-free observational target. Its precipitation is inherited from atmospheric reanalysis forcing, while its soil-moisture and runoff fields depend on land-surface model structure and parameterisation. The corrected outputs may therefore reproduce reference-specific characteristics as well as reduce CMIP6 biases. Improved agreement should consequently be interpreted as increased consistency with ERA5-Land rather than independent evidence of improved observational accuracy or physical realism. Although the ERA5-Land and CMIP6 soil-moisture fields were harmonised to a common upper 10 cm depth, residual uncertainty may remain because the CMIP6 models use different native soil-layer structures and land-

surface parameterisations^{3,4,77}. Independent evaluation against additional precipitation, soil-moisture and runoff or discharge datasets would be required to assess reference sensitivity.

Precipitation, soil moisture and surface runoff were also corrected separately despite their statistical and physical relationships. Independent correction may improve each variable individually while altering cross-variable dependence, temporal lag relationships or shared variability. This is particularly relevant because soil moisture integrates antecedent precipitation and evaporative losses, while surface runoff depends on precipitation, soil-water conditions and delayed land-surface response. The present study did not impose water-balance closure, cross-variable constraints or multivariate dependence preservation. The corrected fields should therefore not be assumed to form a physically closed hydro-climatic system even when each variable shows stronger agreement with ERA5-Land^{74,78-80}. Future work should examine cross-variable correlations, lag relationships, water-balance consistency and multivariate correction approaches.

The remaining monthly-tail biases in Figures 8 and 9 show that improvements in average performance did not extend equally across the upper and lower parts of the distributions. The Q95 and especially Q99 estimates are based on relatively few monthly values during the 84-month testing period and are therefore subject to substantial sampling uncertainty. Nonlinear thresholds and limited representation of high monthly totals in the training record may also contribute to residual upper-tail errors, while average-error loss functions can favour more frequently occurring central values. These results should therefore be interpreted as evidence of remaining bias in monthly upper-tail precipitation and runoff totals and lower-tail soil-moisture conditions, rather than as direct evidence concerning storm intensity, flood peaks, daily extremes or return levels. The contraction of the historical inter-model envelopes in Figure 6 and the tighter clustering in Figures 5 and 7 should be interpreted as reduced historical inter-model dispersion relative to the common ERA5-Land reference, not as complete removal of climate-model uncertainty. Because all models were corrected towards the same dataset, some convergence may arise mechanically and may suppress part of the original model variation. Uncertainty associated with internal variability, model structure, future forcing, reference choice and correction method remains⁸¹⁻⁸³. Uncertainty is also introduced by the selected evaluation metrics and ranking framework. Table 3 shows that different diagnostics can produce different method rankings, particularly for surface runoff. RF produced the lowest RMSE and MAE, whereas U-Net achieved the highest R, NSE and KGE. This difference illustrates why method performance should be assessed using complementary diagnostics rather than a single statistic. TOPSIS reduced reliance on one metric by combining KGE and normalized RMSE, although the resulting ranking remains conditional on the selected criteria and weights. Alternative combinations involving NSE, correlation, bias, regional performance or monthly-tail diagnostics could produce different rankings. Because no weight or criterion sensitivity analysis was conducted, Figure 10 should be interpreted as a structured evaluation under the adopted decision framework rather than a definitive ordering of CMIP6 model quality. Multi-criteria studies similarly show that model rankings vary with the selected variables, metrics and decision rules^{23,52,84,85}.

Preprocessing choices introduce additional uncertainty because regridding, bilinear interpolation, temporal aggregation and land masking can modify the spatial variability of the original fields before correction. Regridding to a common 0.5° grid was necessary for consistent comparison, but bilinear interpolation may smooth local precipitation maxima, sharp soil-moisture gradients and spatial variability in surface runoff, particularly over mountainous, coastal and heterogeneous land-surface regions. Some remaining regional and monthly-tail errors in Figures 4, 8 and 9 may therefore reflect both model bias and information lost during spatial harmonisation. This limitation is particularly relevant when interpreting local correction skill because stronger agreement at the regional or continental scale does not necessarily imply equivalent accuracy at individual catchments or sub-grid scales.

5. Conclusion

Bias correction is a multidimensional and variable-dependent problem because climate-model errors can affect the magnitude, variability, temporal agreement, spatial structure and distributional characteristics of hydro-climatic variables, while improvement in one property does not necessarily lead to comparable improvement in others. The challenge is further complicated by the contrasting statistical behaviour, temporal persistence and spatial organisation of precipitation, surface soil moisture and surface runoff across the terrestrial water cycle. In the present framework, distribution-based correction primarily targeted marginal distributions, RF represented nonlinear model-reference relationships, and the U-Net-based configurations incorporated spatial information from neighbouring grid cells. These methodological differences provide an important basis for understanding why the three correction families showed contrasting strengths across domain-level statistics, spatially distributed grid-cell metrics and distributional diagnostics. Meaningful comparison therefore requires the methods to be assessed under consistent data, spatial, temporal and evaluation conditions using multiple complementary measures of performance. This study addressed this need by comparing QDM, RF and variable-specific U-Net-based configurations for 20 CMIP6 models across Europe using ERA5-Land as a common reference. Overall, the results show that all three approaches improved some characteristics of the raw CMIP6 simulations, but their relative performance varied among hydro-climatic variables and evaluation diagnostics, demonstrating that bias-correction effectiveness cannot be adequately represented by a single metric or a universally preferred correction method.

Among the evaluated configurations, the variable-specific U-Net-based approaches provided the strongest overall multi-metric performance. Relative to the raw CMIP6 ensemble, correlation increased from 0.170 to 0.689 for precipitation, from 0.464 to 0.750 for surface soil moisture and from 0.164 to 0.706 for surface runoff. The U-Net configurations also produced the highest NSE and KGE values and the largest spatial increases in grid-cell correlation. However, this advantage was not uniform across all diagnostics. QDM remained particularly effective in reproducing marginal distributions and selected monthly quantiles, especially for precipitation and soil moisture, while RF reduced several magnitude-based errors but showed less consistent improvements across correlation and efficiency measures. This contrast shows that improved distributional agreement does not necessarily translate into higher temporal correlation or consistently improved performance across individual grid cells. Overall, U-Net provided the most balanced performance across the evaluated dimensions, whereas QDM retained clear advantages when distributional fidelity was the main objective. These findings support selecting bias-correction methods according to the properties most relevant to the intended application.

Bias correction also altered the structure of the CMIP6 ensemble and the relative performance of individual models. The corrected simulations showed narrower historical inter-model dispersion and tighter clustering relative to ERA5-Land, particularly under the U-Net-based configurations. This convergence indicates reduced model-to-model differences relative to the common reference, but it should not be interpreted as elimination of climate-model uncertainty because part of the contraction may arise from adjusting all models towards the same target. Importantly, the TOPSIS analysis showed that bias correction did not benefit all CMIP6 models equally and substantially reorganised their relative rankings, particularly for precipitation and surface soil moisture, whereas runoff rankings were more persistent. This demonstrates that strong raw performance does not necessarily translate into strong post-correction performance, since the correction method can modify model-specific biases to different degrees. Consequently, climate-model selection should not be treated as independent of bias correction when corrected simulations are intended for subsequent hydro-climatic applications. Models should instead be evaluated for the target variable after applying the intended correction approach, while recognising that the resulting ranking remains conditional on the adopted KGE–RMSE criteria and their assigned weights.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in the writing process to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Table Used in this Manuscript

Table 1. CMIP6 models used in this study, including modelling institution, full model name and native spatial resolution.

CMIP6 Model	Institution	Model Abbreviation	Resolution	Reference
ACCESS-CM2	CSIRO-ARCCSS	Australian Community Climate and Earth System Simulator Climate Model Version 2	1.25°×1.875°	Bi et al. (2020) ²⁸
ACCESS-ESM1-5	CSIRO	Australian Community Climate and Earth System Simulator Earth System Model Version 1.5	1.25°×1.875°	Ziehn et al. (2020) ²⁹
BCC-CSM2-MR	BCC	Beijing Climate Centre Climate System Model version 2, Medium Resolution	1.125°×1.125°	Wu et al. (2019) ³⁰
BCC-ESM1	BCC	Beijing Climate Centre Earth System Model version 1	2.8° × 2.8°	Wu et al. (2020) ³¹
CAMS-CSM1-0	CAMS	Chinese Academy of Meteorological Sciences Climate System Model version 1.0	1.125°×1.125°	Rong et al. (2018) ³²
CESM2	NCAR	Community Earth System Model version 2	0.9° × 1.25°	Danabasoglu et al. (2020) ³³
CESM2-WACCM	NCAR	Community Earth System Model version 2 with Whole Atmosphere Community Climate Model	0.9° × 1.25°	Danabasoglu et al. (2020) ³³
CMCC-CM2-HR4	CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici Climate Model 2, High Resolution 4	0.94° × 1.25°	Scoccimarro et al. (2022) ³⁴
CMCC-CM2-SR5	CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici Climate Model 2, Standard Resolution 5	0.94° × 1.25°	Cherchi et al. (2019) ³⁵
CMCC-ESM2	CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici Earth System Model 2	0.94° × 1.25°	Lovato et al. (2022) ³⁶
E3SM-1-0	E3SM-Project	Energy Exascale Earth System Model version 1.0	1.0° × 1.0°	Golaz et al. (2019) ³⁷
EC-Earth3-AerChem	EC-Earth Consortium	EC-Earth3 with interactive aerosols and atmospheric chemistry	0.7° × 0.7°	van Noije et al. (2021) ³⁸
EC-Earth3-CC	EC-Earth Consortium	EC-Earth3 carbon cycle configuration	0.7° × 0.7°	Döscher et al. (2022) ³⁹
KACE-1-0-G	NIMS-KMA	Korean Advanced Community Earth System Model version 1.0, GLOMAP	1.25° × 1.875°	Lee et al. (2020) ⁴⁰
MIROC6	MIROC	Model for Interdisciplinary Research on Climate version 6	1.4° × 1.4°	Tatebe et al. (2019) ⁴¹
MIROC-ES2L	MIROC	MIROC Earth System Model version 2 for Long-term simulations	2.8° × 2.8°	Hajima et al. (2020) ⁴²
MPI-ESM1-2-LR	MPI-M, AWI, DKRZ, DWD	Max Planck Institute Earth System Model version 1.2, Low Resolution	1.87°×1.87°	Mauritsen et al. (2019) ⁴³
MRI-ESM2-0	MRI	Meteorological Research Institute Earth System Model version 2.0	1.12° × 1.12°	Yukimoto et al. (2019) ⁴⁴
NorESM2-MM	NCC	Norwegian Earth System Model version 2, Low atmosphere-medium ocean resolution	1.9° × 2.5°	Seland et al. (2020) ⁴⁵
UKESM1-0-LL	MOHC, NERC, NIMS-KMA, NIWA	UK Earth System Model version 1.0, Low Resolution	1.25° × 1.875°	Sellar et al. (2019) ⁴⁶

Table 2. U-Net bias-correction configurations for precipitation, surface soil moisture and surface runoff.

Variable	Correction design	Model inputs	Prediction target	Temporal structure	Loss function
Precipitation	QDM followed by direct U-Net correction	QDM-corrected precipitation and elevation	ERA5-Land precipitation	Separate seasonal models for DJF, MAM, JJA and SON	MAE + correlation loss + variance loss
Surface soil moisture	Seasonal lagged-anomaly U-Net	CMIP6 soil moisture anomalies	ERA5-Land soil moisture anomaly	Separate DJF, MAM, JJA and SON models using three lagged anomaly fields	MAE
Surface runoff	QDM followed by lagged residual U-Net correction	QDM-corrected runoff anomalies from (t-6) to (t)	Residual anomaly between ERA5-Land and QDM-corrected runoff at month (t)	Separate seasonal models using seven consecutive monthly inputs	MAE

Table 3. Spatial means of grid-cell temporal performance metrics for the raw and bias-corrected CMIP6 multi-model ensemble fields relative to ERA5-Land during the January 2008–December 2014 evaluation period. Higher R, NSE and KGE and lower RMSE, MAE and absolute bias indicate better performance.

Variable	Method	R	RMSE	MAE	Bias	NSE	KGE
Precipitation	Raw CMIP6	0.17	48.04	36.97	2.51	-2.57	-0.058
	QDM corrected	0.23	42.4	32.27	1.04	-0.553	0.21
	RF corrected	0.38	31.3	23.97	-0.28	0.079	0.25
	U-net corrected	0.689	26.93	20.16	1.22	0.38	0.55
Surface Runoff	Raw CMIP6	0.164	14.377	8.831	5.639	-555.774	-11.178
	QDM corrected	0.275	4.035	2.281	-0.052	-14.225	0.245
	RF corrected	0.505	0.350	0.221	-0.024	0.157	0.257
	U-net corrected	0.706	2.425	1.279	-0.3181	0.386	0.571
Soil Moisture	Raw CMIP6	0.464	10.694	9.327	-1.814	-58.926	-0.324
	QDM corrected	0.507	4.518	3.526	0.1802	-0.42	0.431
	RF corrected	0.669	3.338	2.523	0.515	0.397	0.553
	U-net corrected	0.75	3.065	2.304	0.519	0.451	0.598